The Indirect Effects of AI on Professional Work

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Abstract

When customers use AI to discover questions and prepare work, their purchases change the tasks and learning opportunities that reach professional providers. We develop a model in which workers privately own their expertise, finance formal training from future earnings, and pay for diagnostic practice through learning credits in customer fees. With the service demand schedule held fixed, a finite loss of diagnostic practice raises total formal training while reducing the stationary expert population, with replacement strictly below the lost practice. When services and research compete for these people, anticipated machine improvements can reduce current training before adoption, while continued human research sustains expertise through private investment after practice disappears. We derive a threshold comparing service hours released with qualifications lost, which determines eventual human research labor and its frontier growth coefficient. Joint welfare values the public research return separately from the procurement wage, giving a region in which preparation expands research while reducing welfare. Existence and determinacy conditions, human grounding, changing research resources, and explicit career continuation connect these results to the economy's dynamic paths.

Keywords Artificial intelligence, indirect effects, professional services, private human capital, learning contracts, research automation.

JEL classification D24, D61, D83, J24, J31, O33, O41.

1 Introduction

Before a customer brings a software problem to an engineer, part of the eventual service may already have been produced at home, where an AI conversation has helped the customer assemble a working example, trace an error, and decide which remaining question requires professional judgment. Upon opening that file, the engineer encounters a different collection of tasks from the one that would have arrived a year earlier, even if the engineering office continues to use the same tools. Some of the preparation has disappeared, some uncertainty has been resolved, and a question that the customer previously lacked the means to formulate may now be the reason for purchasing help.

Customers who prepare their own work change the occupation through the transactions they bring to it, while people inside the occupation continue to acquire expertise through those transactions. For a junior engineer who once reconstructed a fault from its symptoms, a submitted solution can remove the occasion for understanding why the fault arose, although a file containing the relevant intermediate reasoning and feedback can preserve that occasion. We study how these choices change professional earnings, the private acquisition of expertise, and the research that competes for the same qualified people.

Recent evidence gives the customer side of this process an economic scale and a recognizable set of activities. Google's AI & Economy ATLAS maps Gemini interactions to work and nonwork activities, with more than 86 percent of conversational interactions on Gemini App and AI Mode in its covered April 2026 sample classified as nonwork, including household planning, learning, and practical advice (Iscenko et al., 2026). In their study of ChatGPT use, Chatterji et al. (2025) document practical guidance, information seeking, and writing, activities that can enter the decisions and preparation preceding a professional transaction. These records describe

interactions and requested activities, which we use to locate the customer's contribution to a transaction and the information that can travel with it.

In valuing the work performed on either side of that transaction, we follow Becker (1965) by counting customer time together with purchased inputs, so that a calculation produced at home and an hour supplied by a professional enter the same resource account. For the acquisition of expertise, we follow the individual ownership and earnings logic of Becker (1962) and Ben-Porath (1967), giving a novice a qualification that travels across employers and a formal training choice paid for from future income. Combining those two decisions places a price on the learning occasion contained in a customer file, because a novice who benefits from the occasion can help finance the service through an apprenticeship payment.

Evidence on assisted work gives a reason to distinguish the quality of a completed service from the judgment retained by its producer. In the patent-drafting field experiment of Autor et al. (2026), access to AI and subsequent unaided assessment allow the authors to examine expertise after the tool is removed, while the skill-formation experiment of Shen and Tamkin (2026) follows learning during AI-assisted coding. We build that distinction into the model by measuring learning through the information processed and a subsequent qualification assessment, with ordinary hours contributing only when their feedback bears on the principle being learned.

Within the literature on automation and new tasks, customer preparation relocates a task across the boundary of the professional firm (Acemoglu and Restrepo, 2019), which also changes the questions reaching the organization of knowledge studied by Garicano (2000). Our closest career comparison is Afrouzi et al. (2026), whose economy-wide learning rate creates an interaction across workers and whose discussion distinguishes that mechanism from bilateral training financed through wage concessions. We develop the bilateral case explicitly, deriving the practice credit from the novice's private qualification value and then examining what happens when an institutional payment rule restricts that credit.

With complete contracts, a reduction in diagnostic practice induces privately financed replacement, yet the replacement remains strictly smaller than the lost practice throughout a feasible interval of stationary economies. Expert headcount falls, total formal training rises, and hourly earnings rise wherever expert demand is strictly decreasing, while total expert income has a separate elasticity condition. We also prove that an optional expansion of customer technology raises welfare in the competitive service-and-training economy, and identify a precise restriction on apprentice compensation under which customer preparation has a negative marginal welfare effect.

When engineers can perform both customer work and research, the same qualification market determines the price of generation and validation in the research sector. Customer preparation can release expert time for research while reducing the future entry of qualified people, and the two effects appear together in an exact wage-mediated expression for research output. By specifying complete research methods, we derive how anticipated automation changes current study, how the public return to research differs from its procurement wage, and how private training sustains a human research input after practice vanishes. Idea difficulty, human grounding, resources, and research duration then determine the frontier consequences of that occupational adjustment.

We begin with the customer's file and the information that its preparation preserves, then follow the payment for practice into the novice's career decision. After deriving the stationary response and its welfare consequences, we add fees, risk, and validation to the service market before joining services and research through their common demand for qualified hours. Our final results describe the frontier dynamics of that economy and the economic measurements needed to connect its theoretical mechanisms to observed work.

2 Customers

2.1 Choices

For a recurring population of matters indexed by z and measured by μ_Z, a customer chooses a way of proceeding d from the menu $\mathcal{D}(z, c)$ available at customer capability c. Among these modes are leaving a matter unresolved, completing it at home, purchasing a professional service, and bringing different amounts of preparation to that service. We write $U_d(z, c, f)$ for its expected benefit after customer time, inference resources, and other nonlabor costs, where f describes the provider's tools, while j_d and e_d record the basic and expert clock hours needed to complete it.

By keeping the underlying matters fixed when changing customer capability, we can follow discovery, self-service, information, and preparation through the same population. Discovery changes which circumstances receive attention, self-service improves an outside choice, and preparation changes the work needed after a purchase, while better information changes the conditional benefit of seeking help. For a binary state in which intervention has benefit L with posterior probability p, the option value $(pL - F)_+$ at fee F is convex in p, so a Blackwell refinement of the customer's signal weakly raises the expected value of this purchase option (Blackwell, 1953). Purchased hours then follow the decisions selected at those posteriors.

2.2 Learning credits

Inside the profession, qualification records independent diagnosis, design, and verification in a defined field of software and systems engineering. One unit of expertise belongs to one qualified person, who supplies one expert clock hour per unit of time, and the share of qualified people is $E \in (0, 1)$. Novices supply basic time at opportunity wage $b > 0$, supported by an outside activity that remains available, while experts receive hourly wage w. We work on a region where the basic time requirement is below the novice endowment, so the outside activity fixes b and the scarce expert market determines w.

Each mode also supplies an expected qualification transition m_d to an eligible novice who receives its diagnostic information and feedback, with every realized transition assigned to someone who is still a novice. We denote the private value of acquiring qualification by Δ, whose derivation follows below, and allow the novice to buy the learning opportunity through a payment or a wage concession. Competition then gives the fee and customer choice

$$\begin{aligned} F_d &= b j_d + w e_d - \Delta m_d, \\ d^*(z) &\in \arg\max_{d \in \mathcal{D}(z,c)} \{U_d - b j_d - w e_d + \Delta m_d\}. \end{aligned} \tag{1}$$

For a provider who pays the novice $b j_d$ and the expert $w e_d$, the sum of the customer fee and the novice's learning payment exactly covers those labor payments, so the discount attached to an informative file has an explicit source of finance.

Because the opportunity to learn has a market value, service demand and practice depend on the qualification price as well as the hourly wage. Aggregating the chosen modes gives

$$D_S(w, c, \Delta) = \int e_{d^*(z)} \, \mathrm{d}\mu_Z, \qquad M(w, c, \Delta) = \int m_{d^*(z)} \, \mathrm{d}\mu_Z, \tag{2}$$

with basic demand $J = \int j_{d^*(z)} \, \mathrm{d}\mu_Z < 1 - E$. Research hours will later enter expert clearing alongside D_S, while the service-and-training economy initially allocates all expert hours to professional customers.

To compare customer technologies, we hold a common positive path of provider tools f_t and let customers, workers, and fees adjust to the changed customer path c_t. Direct assistance inside the provider enters U_d, j_d, e_d, m_d through f_t, whose economic role is distinct from the customer work arriving at the office. Task-level cost savings of the type analyzed by Acemoglu (2025) enter through specified tasks and users, and

the prediction-versus-decision distinction in Agrawal et al. (2019) appears when an AI-produced prediction still requires expert interpretation or verification.

When a customer brings a software patch, these choices can increase demand for the engineer who checks an unfamiliar security implication, reduce demand for routine debugging, and alter the learning opportunity available to a novice assigned the file. Changes in the stock of work and in its composition both matter, just as the new activities documented by Autor et al. (2024) change the content of professional employment over time. Our mode description retains those margins through the purchased file and the information passed to its recipient.

3 Learning

3.1 Diagnostic information

When someone reconstructs a fault, the value of the exercise for later judgment depends on whether the reconstruction reveals a principle used in that judgment. We illustrate this dependence with a foundation B and a target principle T, each taking the values -1 and 1 with equal probability and independently of the other. By processing a foundational observation, the learner receives $S = B\eta$, while an explanation gives $Y = BT\epsilon$, where the independent noises have mean reliabilities r_B and r_A in $[0, 1]$.

Without the foundational observation, averaging over the unknown foundation leaves both target values equally likely, while a learner who processes the foundation and the explanation has

$$\mathbb{E}[T \mid S, Y] = r_B r_A S Y, \qquad \mathcal{R}(S, Y) = 1 - r_B^2 r_A^2, \tag{3}$$

where $\mathcal{R}$ is the mean squared error of a subsequent unaided judgment. Knowing the foundation in advance gives risk $1 - r_A^2$, so the same explanation produces different retained ability according to what the recipient already understands, with the calculation proved in Proposition A.1.

During customer preparation, a lost reconstruction occasion reduces access to S, while a submitted explanation that preserves both signals retains the same learning value. If access to the foundational signal has probability p, independent of the principle and signal noise, expected learning is $p r_B^2 r_A^2$, giving a measurable comparison between the frequency of diagnostic exposure and the quality of what is processed. For explanations that directly reveal the target through $T\epsilon$, the foundational observation ceases to be required, which also gives a model of instruction that replaces the removed experience.

Across several principles, the same point has a covariance formulation. For a prior $\theta \sim N(\bar{\theta}, \Sigma)$, an exercise $Z = u^\top \theta + \varepsilon$ with independent Gaussian noise $\varepsilon \sim N(0, \sigma^2)$ and $\sigma^2 > 0$ reduces the unaided error for target $v^\top \theta$ by

$$\frac{(v^\top \Sigma u)^2}{\sigma^2 + u^\top \Sigma u}. \tag{4}$$

Through the conditional covariance calculation proved in Corollary A.2, which has the rank-one algebra associated with Sherman and Morrison (1950), practice transfers exactly through its prior covariance with the target, giving zero transfer when that overlap vanishes.

3.2 Qualification and exposure

To connect this information experiment to occupational entry, we specialize to a qualification margin at which a fully revealing diagnostic occasion permits a novice to pass an unaided assessment, while formal instruction supplies another way to reach that margin. Case content determines which occasions reveal the relevant principle, and retained reconstruction and feedback determine which ones reach a learner. With $\mathcal{D}$ denoting diagnostic cases and $\mathcal{R}(c)$ the reconstruction occasions retained by novices, a purchased case density $n(z, c)$ gives

$$M(c) = \int_{\mathcal{R}(c) \cap \mathcal{D}} n(z, c)\, \mathrm{d}z \tag{5}$$

in the specialization where each retained diagnostic occasion produces one transition, so a removed hour changes learning only through its diagnostic content.

Learning by doing in Arrow (1962) supplies the economic interpretation of this joint production of output and future ability, while the cognition model of Acemoglu et al. (2026) motivates examining how access to AI changes the acquisition of underlying knowledge. Evidence from Brynjolfsson et al. (2025) shows how assistance can transmit useful practices during customer support, and the task variation studied by Dell'Acqua et al. (2026) gives a reason to allow assistance and reliability to vary across files. Our diagnostic measure follows the later assessment, the intermediate information processed, and the task to which that information transfers.

In an empirical application, M can be constructed from the arrival of cases, their diagnostic classification, the recipient's actual exposure, and verified qualification transitions, while E counts people who pass the specified assessment. Theoretical results below concern a change in those objects, whose units are transitions per unit of time and a population share, so the comparison can be stated without choosing an arbitrary scale for a customer capability index.

Under targeted allocation, the information experiment defines which opportunities enter M and which recipients cross the qualification margin, while the career state records ownership of that qualification. When diagnostic files instead contact people throughout the occupation, a contact reaches an eligible novice with probability $1 - E$, giving the alternative acquisition technology

$$M = (1 - E)m(c), \qquad \Pr(\text{acquisition by } t) = 1 - \exp\left\{-\int_0^t m(c_s)\,\mathrm{d}s\right\} \tag{6}$$

for an initially novice cohort without obsolescence or formal instruction. Repeated exposure has a decreasing acquisition return because the first revealing contact already supplies the principle, and the finite incomplete-replacement result survives this saturation, as proved in Appendix C.3. Our targeted allocation supports the contract benchmark below, while random contact adds the social effect of removing a potential recipient from future learning.

4 Careers

4.1 Private investment

When a novice can carry qualification to another employer, its value is the difference $\Delta = V_E - V_N$ between the discounted values of expert and novice status. Qualification becomes obsolete at rate $\delta > 0$, returning its owner to novice status, while formal study supplies a transition intensity $x \geq 0$ at resource cost $\kappa x^2/2$ with $\kappa > 0$. By paying this cost to an independently supplied teaching sector, whose resources are measured in the numeraire, the novice finances study alongside diagnostic placements, with borrowing against future earnings available for both purchases.

Because diagnostic practice arrives at rate $g = M/(1 - E)$, a payment a per successful transition gives the novice expected capital gain $g(\Delta - a)$. Under complete pricing $a = \Delta$, the practice seller receives its private value, while the novice chooses formal intensity from the remaining investment problem. With discount rate $\rho > 0$ and private qualification value $\Delta \geq 0$, the individual value equations in Appendix B give

$$\begin{aligned} x &= \frac{\Delta}{\kappa}, \\ \dot{E} &= M + (1 - E)x - \delta E, \\ \dot{\Delta} &= (\rho + \delta)\Delta - (w - b) + \frac{\Delta^2}{2\kappa}. \end{aligned} \tag{7}$$

Learning opportunities and formal instruction are thus financed by the worker who acquires the asset, whose mobility preserves both the future earnings and the obligation to repay the original financing.

4.2 Practice replacement

To obtain a finite comparison with an explicit sign, we fix a purchased diagnostic portfolio whose selected modes retain strict purchase margins over the comparison and an elastic portfolio of nondiagnostic expert services. Changing the diagnostic content of the first portfolio changes M while preserving its expert requirements, so the inverse wage schedule $w(E) > b$ is continuous, nonincreasing, and independent of M and Δ on the relevant region. We use an interval $[E_-, E_+] \subset (0, 1)$ where novice time remains feasible and where the qualification drift is positive at E_- for the smallest practice flow and negative at E_+ for the largest flow.

For these economies, writing $R = \rho + \delta$, stationarity gives

$$x(E) = \sqrt{R^2 + 2[w(E) - b]/\kappa} - R, \qquad M + (1 - E)x(E) = \delta E. \tag{8}$$

Because the first expression decreases weakly with the expert population and the number of novices decreases strictly, the total formal transition flow $(1-E)x(E)$ decreases strictly with E. Together with this monotonicity, the stated endpoint conditions give one stationary expert population for every practice flow in the interval.

Proposition C.1 proves that every finite practice loss $M^1 < M^0$ in this region produces

$$E^1 < E^0, \qquad 0 < T^1 - T^0 < M^0 - M^1, \qquad T^i = (1 - E^i)x^i, \tag{9}$$

while formal resource expenditure $\kappa(1 - E)x^2/2$ rises strictly. Whenever the inverse wage falls strictly over the traversed expert interval, the smaller expert population raises w, the skill premium $w - b$, the private asset value Δ, and individual study intensity x, giving an equilibrium response to the lost practice that can be observed separately in prices, people, and training payments.

For differentiable wage schedules, the replacement rate has the exact expression

$$\Lambda = \delta + x - \frac{(1 - E)w_E}{\kappa(R + x)} > \delta, \qquad E_M = \frac{1}{\Lambda}, \qquad -T_M = 1 - \frac{\delta}{\Lambda} \in (0, 1). \tag{10}$$

Because κ prices formal training and the labor market determines the wage response to a shrinking expert population, the worker's optimal response determines this rate. Full replacement would preserve the old headcount and wage, leaving the original private study incentive unchanged, whereas an increase in individual study intensity requires a change in the career return.

When customer discovery also raises the inverse wage at fixed headcount, the same stationary system yields

$$E_c = \frac{M_c + (1 - E)w_c/[\kappa(R + x)]}{\Lambda}. \tag{11}$$

For $w_c \geq 0$, expertise declines precisely when the diagnostic loss $-M_c$ exceeds the formal entry response $(1 - E)w_c/[\kappa(R + x)]$ encouraged by the demand shift, giving a structural threshold in objects tied to cases, wages, and study costs. Within that region, the induced scarcity wage reinforces the direct wage increase and total training rises, while the finite result in (9) applies when the demand schedule is held fixed.

4.3 Earnings

Hourly earnings and the number of their recipients enter total expert income separately, since

$$\frac{\mathrm{d}(Ew)}{\mathrm{d}M} = \frac{w + Ew_E}{\Lambda}. \tag{12}$$

With isoelastic expert demand $D_S = Cw^{-\varepsilon}$, the wage bill is proportional to $E^{1-1/\varepsilon}$, so a practice loss reduces total expert income when $\varepsilon > 1$ even as hourly earnings rise. Novice cash income includes the placement and tuition payments, while its stationary lifetime value is $V_N = b/\rho + \Delta^2/(2\kappa\rho)$ and expert value is $V_E = V_N + \Delta$,

giving explicit accounts for the distributional objects emphasized in the knowledge-allocation analysis of Garicano and Rossi-Hansberg (2006).

Over time, E is inherited and Δ is the value of future career opportunities, so an announced customer path can affect current study through future wages before current practice changes. Proposition G.3 expresses the difference between two bounded global career-value paths as a positively weighted integral of future wage differences, while its finite-horizon counterpart uses common terminal qualification values. Customer acceleration enters that result through the equilibrium wage paths generated by purchases and training, with the general sensitivity system in (45) retaining both the service-demand and practice responses along the complete path.

5 Contracts

5.1 Service welfare

For the competitive service-and-training economy, all payments for labor and practice can be added across customers, providers, and workers, while the resources used in teaching remain a real cost. Writing $I = (1-E)x$ for the aggregate formal transition flow, the discounted resource objective is

$$\mathcal{V} = \int_0^\infty e^{-\rho t} \left[b(1-E) + \int (U_d - b j_d)\, \mathrm{d}\mu_Z - \frac{\kappa I^2}{2(1-E)} \right] \mathrm{d}t, \tag{13}$$

subject to qualification accumulation, expert feasibility $D_S \le E$, and basic feasibility $J \le 1 - E$. By valuing novice time in its outside use through $b(1 - E)$ and subtracting bJ, we account for the outside output displaced by professional work.

Because the training cost is a convex perspective and the remaining constraints are affine in allocation shares and the expert population, this problem is concave. Proposition D.1 verifies that a feasible competitive allocation with complete practice pricing, slack basic time, and a discounted private value tending to zero attains its global maximum, with social costate $\lambda = \Delta$. Under those prices, the customer's learning credit equals the social value of the diagnostic transition, while the privately chosen study intensity supplies the same formal investment as the resource allocation problem.

When an optional improvement preserves all previously available customer modes and adds new ones, every old allocation remains feasible, so discounted welfare weakly rises under the conditions of this implementation result. Expertise can fall during this efficient adjustment when the new customer choices save enough current resources to pay for the lost practice at its full value. By following entire feasible paths, the welfare comparison includes both the current service and the future professional population.

5.2 Payment restrictions

Institutional restrictions change the preparation choice for a negligible class of cases around that benchmark, each of which can be prepared by the customer at cost $r_C \ge 0$, saving one basic hour and removing a diagnostic transition probability $m > 0$. With a positive minimum cash compensation $s \in (0, b)$ and a prohibition on separately charging the apprentice for the placement, the learning payment is capped at $b - s$. When that cap binds, customer preparation is chosen even though its marginal welfare effect is negative exactly when

$$b - s < b - r_C < \Delta m. \tag{14}$$

Proposition E.1 proves the interval and its feasibility, including the symbolic case $b = 1$, $s = 1/2$, $r_C = 3/8$, and $\Delta m = 3/4$, where the privately perceived saving is positive and the resource gain net of lost qualification value is negative.

Once the restriction applies to a positive share of the whole economy, the private asset value and the social value of expertise must each be solved with the altered allocation, and the respective preparation gains are

$b - r_C - am$ and $b - r_C - \lambda m$. Their difference $(\lambda - a)m$ identifies the missing payment whose size determines the distortion, while the local result above supplies a signed case from an explicit institution. In this way, the financing arrangement determines whether customer preparation is guided by the future value of what its recipient would have learned.

6 Service quality

6.1 Fees and reliability

For services whose quality customers find difficult to assess, the fee also purchases a judgment about reliability, as the credence-goods analysis of Dulleck and Kerschbamer (2006) emphasizes. In the market-power extension, engineering firms own fixed franchises serving separate customer groups, with a common technology and demand elasticity and purchases independent of other firms' prices. Franchise ownership determines who receives provider profits, while the portable worker qualification determines who supplies expert hours to either services or research. Their aggregate nondiagnostic demand is $Q = N(c)P^{-\varepsilon}$ with $\varepsilon > 1$, where P is the fee plus the customer's certainty-equivalent loss from an unresolved error. Each case requires $e(c) > 0$ expert generation hours, $v \geq 0$ expert validation hours, and other resources $k(c)$, with error probability $z = \pi(c)e^{-v}$ and loss $L > 0$.

With constant absolute risk aversion $\gamma > 0$, the loss has certainty-equivalent cost $r_\gamma(z) = \gamma^{-1}\log[1+z(e^{\gamma L}-1)]$, whose risk-neutral limit is Lz. When a provider chooses both validation and a fee per completed case, it minimizes the generalized unit cost and then applies its demand markup, giving

$$\begin{aligned} \mathcal{T}(w,c) &= \min_{v\geq 0}\{w[e(c)+v] + k(c) + r_\gamma(\pi(c)e^{-v})\}, \\ P &= \mathfrak{m}_f\mathcal{T}, \qquad \mathfrak{m}_f = \frac{\varepsilon}{\varepsilon-1}, \qquad F = P - r_\gamma. \end{aligned} \tag{15}$$

Lemma F.1 derives this choice and proves that the resulting expert demand $D_S = \mathcal{T}_w N(c)(\mathfrak{m}_f\mathcal{T})^{-\varepsilon}$ is strictly decreasing in the wage, including the points at which validation begins or ends.

6.2 Validation and value

With risk-neutral customers and $\pi(c) > 0$, optimal validation is $v^* = [\log(L\pi(c)/w)]_+$, so a more reliable customer file reduces required checking at the given wage, while a larger loss raises the return to expert review. When a prepared file saves generation time, the final service cost still depends on checking and on the wage of the experts performing it, so the provider's quality choice connects customer reliability to the demand for professional judgment.

With market power, an additional expert hour has a current service value exceeding its wage by the amount associated with the extra cases it supports. For E_S hours allocated to this portfolio and service surplus $\mathcal{S}$ evaluated at the fee-and-validation equilibrium, the exact local identity is

$$\frac{\mathrm{d}\mathcal{S}}{\mathrm{d}E_S} = w + (\mathfrak{m}_f - 1)\mathcal{T}\frac{\mathrm{d}Q}{\mathrm{d}E_S} > w, \tag{16}$$

as proved in Appendix F. Provider profits accrue to the fixed franchise owners and cancel as transfers in the aggregate resource account, while the markup wedge changes the social return to qualified labor, whose comparison with private career returns now includes this additional term.

7 Research

7.1 Complete methods

For software and systems engineers, the assessed ability used in customer projects also enters the generation and evaluation of research in the same technical field. We give a research fund a fixed expenditure flow $B_R > 0$, financed by lump-sum contributions from an independently endowed numeraire sector, and a mandate to purchase the greatest number of completed validated research packets. Within the research-organization perspective of Jones (2025), this budget directs attention to the cost of a completed result and the allocation of inputs among its stages.

We measure the frontier by a cardinal productivity index $A \geq A_0 > 0$ and connect it to the customer interface through $c(A) = \kappa_c \log(A/A_0)$ with $\kappa_c > 0$, while current research uses capability $r = A$. Completing a human packet requires generation hours $A^{-\alpha}$ and validation hours $v_R A^{-\nu}$, with $v_R > 0$, and each researcher hour uses $\xi \geq 0$ computation units at price $q > 0$. Autonomous completion requires generation computation $A^{-\mu}$ and machine-validation computation $m_R A^{-\zeta}$ with $m_R > 0$, giving

$$\begin{aligned} a_H(A) &= A^{-\alpha} + v_R A^{-\nu}, \quad k_H = a_H(A)(w + q\xi), \\ a_M(A) &= A^{-\mu} + m_R A^{-\zeta}, \quad k_M = q a_M(A). \end{aligned} \tag{17}$$

All four elasticities are nonnegative, and complete autonomy becomes available where both stages can be performed with zero human hours. For customer-use comparisons, an intervention shifts the interface to $c_\vartheta(A)$ at the same frontier, while research requirements retain their common specification.

Given a human packet flow z_H, validation uses $v_R A^{-\nu} z_H$ hours, so its capacity depends on allocated expertise and the productivity of that expertise. For the machine route, the packet includes the computation required to evaluate and integrate its output, following the research stages described in OpenAI (2026). When a tool accelerates generation while checking remains costly, its gain enters one component of a complete production method whose other component continues to require resources.

7.2 Allocation and earnings

With L_R expert hours in research and Q_M autonomous computation, the wage and budget satisfy

$$E = D_S(w, c(A), \Delta) + L_R, \qquad (w + q\xi)L_R + qQ_M = B_R. \tag{18}$$

For the wage w_0 that clears services alone and the wage w_1 that clears services together with human research, strict demand monotonicity gives $w_1 > w_0$. Comparing these wages with $\overline{w} = q a_M/a_H - q\xi$ determines whether the fund buys human packets, machine packets, or both, as proved in Proposition G.1.

When $w_1 \leq \overline{w}$, human research exhausts the fund, whereas $w_0 \geq \overline{w}$ gives an autonomous allocation. In the intermediate region $w_0 < \overline{w} < w_1$, equal packet costs fix $w = \overline{w}$, and the labor market assigns $L_R = E - D_S(\overline{w}, c, \Delta)$ hours to human research while the remaining budget purchases autonomous packets. At a fixed frontier, changes in the expert population then change the human share of research at a constant wage.

For idea difficulty $\phi > 0$, completed packets advance the frontier according to

$$\dot{A} = A^{-\phi}(z_H + z_M) = A^{-\phi} \frac{B_R}{\min\{k_H, k_M\}}, \tag{19}$$

with the minimum taken over feasible methods. Following the research-productivity concern in Jones (1995) and the evidence assembled by Bloom et al. (2020), an absolute frontier increment requires more packets as A grows, so improvements in research tools compete with the increasing difficulty of the next increment.

At a strict human corner, we write $h = w + q\xi$ and $\mathcal{J} = -D_{S,w} + B_R/h^2 > 0$, so a customer perturbation at a common frontier gives

$$\mathrm{d}\log \dot{A} = \frac{\mathrm{d}E - \mathrm{d}D_S|_w}{h\mathcal{J}}, \qquad \mathrm{d}D_S|_w = D_{S,c}\,\mathrm{d}c + D_{S,\Delta}\,\mathrm{d}\Delta. \tag{20}$$

Loss of qualified people and expansion of service demand both raise the research wage and reduce the packets bought by the fund, while a release of service hours enters with the opposite sign. Proposition G.2 also gives a finite ordering when service-demand schedules and expert populations are ordered in the matched-frontier comparison.

Research expenditure pays wL_R to workers and $q(\xi L_R + Q_M)$ to computation suppliers, giving labor's budget share as $w/(w + q\xi)$ in the human region and zero in the autonomous region. At fixed (A, E, c, Δ), access to the cheaper machine method weakly lowers the wage, skill premium, and total expert earnings, as proved in Proposition G.4, while compute receipts include the resources used to provide computation. In the scenarios of Korinek et al. (2026), wages, labor reallocation, and income shares are distinct outcomes of adoption and autonomy, and our expenditure share follows from the specified procurement budget and factor payments.

7.3 Private paths

Together, (7), (18), and (19) form a joint system in inherited states E, A and forward value Δ. Research wages change private training through future skill premia, customer practice changes the people available for research, and the resulting frontier changes customer capability. In the mixed and autonomous regions, machine packet cost fixes current frontier output, giving $F_E = 0$ at a common frontier, where $F = \dot{A}$, while earnings and service welfare continue to depend on expertise.

To select a path, we specify inherited E, A and a terminal career value supplied by a continuation occupation. Proposition H.1 proves existence and uniqueness on a compact feasible region when demand decreases uniformly with the wage and the horizon is short enough to bound the feedback between state paths and private values. Its construction integrates the states forward and the private value backward, with an explicit contraction bound that permits endogenous crossings between human, mixed, and autonomous allocations.

For longer intervals, Proposition H.2 gives a scalar condition on the derivative of the terminal-value residual that identifies a locally unique path, including transversal changes in method availability. In the fixed-frontier diagnostic portfolio, Proposition H.3 proves that the stationary career system has one stable and one unstable direction, with a locally unique convergent private value for each nearby inherited expert population. These results supply the path conditions used in the comparisons below, while the service welfare theorem verifies the optimality of its specified competitive candidate.

7.4 Anticipated automation

When both methods operate, the wage offered to researchers follows their relative complete-method productivities. With $\epsilon_H = -Aa'_H/a_H$ and $\epsilon_M = -Aa'_M/a_M$, equal packet costs imply

$$\frac{\mathrm{d}\overline{w}}{\mathrm{d}\log A} = (\overline{w} + q\xi)(\epsilon_H - \epsilon_M). \tag{21}$$

Reducing machine requirements changes this wage schedule, and a frontier advance lowers the mixed wage wherever machine productivity has the larger elasticity.

Proposition H.4 considers an announced reduction in machine requirements that will be adopted at a future date, with common inherited states and terminal qualification values and both economies remaining in the mixed region. When the improved machine method's elasticity is at least the human method's elasticity over the relevant frontier interval, the two frontier and wage paths agree before adoption, while the improved method subsequently advances the frontier faster and offers a weakly lower wage. Applying the positive-weight career-value formula lowers current Δ, individual study x, and current aggregate training whenever the future wage reduction is strict over a positive interval.

Because the current frontier and wage are still the same at the announcement date, this reduction in training is an anticipation effect generated by future research automation. At a frozen frontier, Proposition H.5

gives a companion finite comparison for first access to a machine route, allowing future expert scarcity to respond while proving that a pointwise reduction in the wage schedule lowers current investment. Once the frontier itself responds, the mixed-regime proposition supplies the joint example and the general value formula retains the complete future wage difference.

8 Research welfare

8.1 Public returns

For the joint economy, the research fund continues to buy packets under its expenditure mandate, while resource welfare values the services and future technology produced by those purchases. At the selected competitive continuation, current resources are

$$G = b(1-E) + \int (U_d - b j_d)\,\mathrm{d}\mu_Z - \frac{\kappa(1-E)x^2}{2} - q(\xi L_R + Q_M). \tag{22}$$

Customer payments, wages, and the fund's lump-sum contribution cancel as transfers, while the last term counts the compute resources and the common labor constraint counts the service opportunities forgone in research. By adding a specified terminal resource value to the discounted integral of G, we define the continuation value $V(t, E, A)$.

We write $\lambda = V_E$ for the social value of an additional qualified person and $\Theta = V_A$ for the social value of a frontier increment. Along a smooth selected equilibrium continuation, Proposition I.1 derives

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(\lambda - \Delta) &= (\rho - f_E)(\lambda - \Delta) - \Gamma_E, \\ \Gamma_E &= L_R w_E + \Theta F_E, \qquad f = \dot{E}, \quad F = \dot{A}. \end{aligned} \tag{23}$$

These derivatives follow the continuation's response to the inherited expert population at a fixed frontier, so the equation values both the fund's wage adjustment and the innovation that an additional expert supports.

In a strict human research region, the forcing term becomes

$$\Gamma_E = (\Pi - h)(L_R)_E, \qquad \Pi = \Theta A^{-\phi}/a_H, \qquad h = w + q\xi. \tag{24}$$

Here Π is the gross social return to a research hour and h is its wage-plus-compute cost. When terminal qualification values agree and future human research has $\Pi \geq h$ with positive research-labor response, the positive-kernel solution gives $\lambda \geq \Delta$, with strictness whenever the innovation wedge is strictly positive over a future interval. Using an explicit public research dividend, the appendix constructs this case and proves that a small subsidy to formal qualification then has positive marginal welfare.

8.2 Released expert time

For one customer case, preparation can save j basic hours and e expert hours, consume resources r_C, and remove qualification probability m. At a strict human research corner of the diagnostic portfolio, a fraction

$$\chi_R = \frac{B_R}{h^2 \mathcal{J}} \in (0, 1)$$

of the released expert time enters human research, while the remainder supports expanded professional services as the wage falls. For a short unannounced intervention followed by the same continuation rule, Proposition I.3 gives the private gain and marginal resource value

$$\begin{aligned} P &= bj - r_C + we - \Delta m, \\ \mathcal{W} &= P - (\lambda - \Delta)m + (\Pi - h)\chi_R e. \end{aligned} \tag{25}$$

Research's contribution enters through its social return net of input costs, while the lost learning opportunity carries the social qualification value.

When $K_C = (\lambda - \Delta)m - (\Pi - h)\chi_R e > 0$, every preparation saving with $0 < P < K_C$ is privately attractive and has negative marginal welfare, even though research hours increase by $\chi_R e > 0$. In the appendix, a feasible symbolic construction with nonnegative preparation costs determines both the direction of research output and the direction of total welfare inside the comparison. For randomly allocated learning contacts, the qualification-gap equation also includes the direct saturation term $-\Delta m(c)$, reflecting the reduction in eligible recipients as expertise expands.

9 Growth

9.1 Practice and research

Human research expenditure supplies a force sustaining expertise after diagnostic practice has disappeared, because clearing requires $w+q\xi \geq B_R/E$. When the expert population becomes small, the earnings equation limits how long training can remain below replacement. Proposition J.1 turns that observation into a strictly positive population bound along global feasible human-research career paths, allowing M to fall all the way to zero.

For a bounded professional portfolio, the service-demand envelope decreases to zero as wages rise, so the derived population floor also gives an upper wage bound. Formal investment now supplies the human input required for the growth comparison, while customer practice can continue to decline as $c(A)$ rises. Research quantity responds through its eventual allocation of expert time, which has an exact finite comparison once the customer's effects on remaining tasks settle.

We denote the eventual diagnostic flows by $\overline{M}_i \in [0, \delta)$ and service schedules by $D_i(w)$, with $i = 0, 1$ indexing two customer environments. Capability may keep increasing after these task requirements have reached their limiting configuration, including complete removal of diagnostic practice in environment 1. At stationary wages, occupational supply is $[\overline{M}_i + x(w)]/[\delta + x(w)]$, and Proposition J.5 proves

$$L_1 \geq L_0 \quad \Longleftrightarrow \quad D_0(w_0) - D_1(w_0) \geq \frac{\overline{M}_0 - \overline{M}_1}{\delta + x(w_0)}. \tag{26}$$

On the right side, the ratio is the loss of qualified people at the old training incentive, while the left side of its comparison is the professional time released at the old wage.

With the service schedule fixed, practice loss lowers eventual human research labor, while sufficiently large service savings raise it. Proposition J.4 proves convergence of every global feasible positive-value career tail in this specialization, and the local saddle result supplies such paths near the stationary point. If $a_H(A) \sim a_\infty A^{-p_H}$, the research coefficient is $K_i = L_i/a_\infty$, so below the critical exponent

$$A_i(t)^{1+\phi-p_H} \sim (1 + \phi - p_H)K_i t, \tag{27}$$

while at $p_H = \phi + 1$ the proportional growth rate tends to K_i. Thus the customer threshold determines a long-run frontier coefficient or, at the critical technology, the exponential growth rate.

9.2 Automation and grounding

When research requires generation and validation, the slower-improving stage determines complete-method productivity, giving $p_H = \min\{\alpha, \nu\}$ and $p_M = \min\{\mu, \zeta\}$. Representing each stage by a positive input requirement supplies a concrete counterpart to the weak-link mechanisms analyzed by Jones and Tonetti (2026). For persistent human research, the population and wage bounds just derived give frontier drift of order $A^{p_H-\phi}$. More generally, with the effective human price bounded above and away from zero, two continuously available methods give exponent $p = \max\{p_H, p_M\}$ and drift of order $A^{p-\phi}$, yielding power growth below $p = \phi + 1$, exponential order at equality, and finite-time production divergence above it.

Complete autonomous validation gives a separate economic outcome, since machine packet cost fixes frontier output independently of human expertise in the machine and interior mixed regimes. Human knowledge continues to affect services, career earnings, and welfare, while this particular local research channel has zero derivative. When the same fixed fund can substitute between two equal-cost complete methods, extra experts change the method shares.

To retain human involvement in machine research, Proposition J.7 gives each machine packet a grounding requirement $a_G(A) > 0$ expert hours, so its cost becomes $qa_M + wa_G$. In the grounded corner, frontier output increases strictly with expertise at a fixed frontier. With $a_G = g_G A^{-\chi}$ and $\chi \leq p_M$, procurement and private training maintain the expert population even after practice disappears, and the complete-route exponent becomes χ. With a grounding stage satisfying $\chi \leq \phi + 1 < p_M$, frontier paths are global where fully autonomous validation would produce a finite endpoint, while equal-cost mixing can still absorb a local change in expertise through method shares.

For the distinction between levels and proportional growth, $\dot{A} = kA^{\beta}$ gives $\ddot{A} = \beta k^2 A^{2\beta-1}$ and $\dot{g}_A = (\beta-1)g_A^2$. Increasing absolute increments coexist with a falling proportional growth rate when $0 < \beta < 1$, and the full human research law adds the wage term

$$\frac{\dot{g}_A}{g_A} = (\epsilon_H - \phi - 1)g_A - \frac{\dot{w}}{w + q\xi}. \tag{28}$$

Research automation and idea difficulty enter the first term through the feedback studied by Davidson et al. (2026), while our customer and career mechanism determines the wage response. In the wider economic account of Jones (2026), the resources available to research also shape the consequences of its productivity.

9.3 Research resources

When research expenditure and compute prices change over time, the autonomous frontier follows the exact resource clock

$$\int_{A_0}^{A(t)} u^{\phi} a_M(u)\,\mathrm{d}u = \int_0^t \frac{B_R(s)}{q(s)}\,\mathrm{d}s. \tag{29}$$

Proposition J.9 determines boundedness, global growth, and finite arrival from these two integrals. For frontier-indexed resources $B_R(A) \sim B_0 A^{\beta_B}$ and $q(A) \sim q_0 A^{\beta_q}$, the autonomous divergence threshold becomes $p_M + \beta_B - \beta_q > \phi + 1$, while human research remains subject to its physical bound $L_R \leq E$. Increasing the budget can bid up the human wage when the available expertise is already employed, so the human response follows both procurement and career entry.

9.4 Research time

As a research tool is adopted, effective capability r can approach the frontier with absorption time $\ell > 0$. For fixed resource scale $k > 0$ and complete-method productivity elasticity $\eta \geq 0$, the system

$$\dot{A} = kr^{\eta}A^{-\phi}, \qquad \dot{r} = (A - r)/\ell, \qquad A(0) = r(0) > 0 \tag{30}$$

has the classification proved in Proposition J.10. At $\eta = \phi + 1$, its limiting growth rate solves $g(1 + \ell g)^{\eta} = k$, while higher elasticities produce a finite endpoint and lower elasticities give power asymptotics.

If each research cycle requires a minimum duration $\tau > 0$, the production law becomes $\dot{A}(t) = kA(t - \tau)^{\eta}A(t)^{-\phi}$. Transforming to $X = A^{1+\phi}$ places the equation within the positive delayed-power class studied by Appleby et al. (2013), and Proposition J.11 supplies a direct proof of its three growth orders. Below the threshold the same power asymptotic survives, at equality the exponential rate solves $ge^{(1+\phi)g\tau} = k$, and above it

$$\lim_{t\to\infty} \frac{\log\log A(t)}{t} = \frac{1}{\tau}\log\frac{\eta}{1+\phi}. \tag{31}$$

Every finite date then has a finite frontier, while the positive double-logarithmic rate quantifies the remaining superexponential growth.

9.5 Career continuation

For the instantaneous autonomous technology with a finite arrival clock, we specify a continuation that gives the endpoint an economic meaning. At completion of the public software program, the fund retires and its algorithmic library becomes available, while qualified engineers retain their credential for residual implementation and verification. In the continuation activity, each expert hour produces $w_\infty > b$ units of the numeraire, novice productivity remains b, and the same teaching technology and obsolescence continue with zero diagnostic practice.

Competitive production then gives the explicit continuation qualification value

$$\Delta_\infty = \kappa \left[\sqrt{(\rho+\delta)^2 + 2(w_\infty - b)/\kappa} - (\rho+\delta) \right], \tag{32}$$

and expertise follows a global linear accumulation law with study intensity Δ_∞/κ. Proposition H.6 derives these continuation values, and Proposition H.7 solves the precompletion career problem using this payoff and proves convergence of finite-cap economies that all use the same continuation activity. Once the program is completed, an absorbing research state replaces the infinite productivity coordinate, while wages, training, and lifetime values continue in calendar time.

10 Conclusion

Customers who use AI change the work that reaches professionals, and the future consequences depend on the information and practice retained in those transactions. By following a file into a novice's qualification, its price into an apprenticeship payment, and that qualification into future earnings, we obtain a private mechanism through which a loss of diagnostic practice induces more formal training and a smaller stationary expert population. Hourly scarcity earnings, individual career value, and the total income received by experts then follow distinct equations.

When learning rights can be priced completely, customer choices account for the future qualification value of the tasks they displace, and optional improvements raise welfare in the competitive service-and-training economy. Restrictions on those payments and markups in professional fees introduce specific differences between private and social returns, whose consequences can be traced to an explicit contract or service-demand condition. For measurement, the relevant observations are the tasks completed, the information received, retained unaided judgment, training payments, and the number of people acquiring qualification.

When the same expertise supplies research, customer demand and career entry jointly determine the wage and the work purchased by a research budget. Anticipated machine improvements can lower training before adoption, while formal investment sustains expertise when human research remains valuable after practice disappears. For this customer effect, the long-run research threshold and welfare value depend on innovation and learning together, with complete autonomy, residual human grounding, and research duration giving different consequences for the same occupational change.

A Learning

A.1 Processed information

Individual unaided judgment concerns a target principle $T \in \{-1, 1\}$, and we represent the foundation needed to interpret a relational explanation by an independent symmetric sign $B \in \{-1, 1\}$, with both signs initially having probability one half. When the person processes a foundational observation and an explanation, the received signals are

$$S = B\eta, \qquad Y = BT\epsilon, \qquad \Pr(\eta = 1) = \frac{1 + r_B}{2}, \qquad \Pr(\epsilon = 1) = \frac{1 + r_A}{2}, \tag{33}$$

where $r_B, r_A \in [0, 1]$ and the two noises are mutually independent and independent of (B, T). We evaluate retained judgment by the minimum expected squared error in estimating T from the information that the person has processed, so this measurement concerns an unaided decision made after the initial service has been completed.

Proposition A.1 (Foundations and interpretable explanations)**.** *Processing only the explanation Y in* (33) *gives posterior mean zero and Bayes risk one, whereas processing both signals gives*

$$\mathbb{E}[T \mid S, Y] = r_B r_A SY, \qquad \mathcal{R}(S, Y) = 1 - r_B^2 r_A^2. \tag{34}$$

When the foundation B is already known, the risk after processing the explanation is $1 - r_A^2$. If receipt of S is independent of the principles and noises and occurs with probability p, while the explanation is available in either event, the ex ante risk reduction is $pr_B^2 r_A^2$. Observations concerning an independent foundation have zero transfer to this target, and customer materials that convey the same processed pair (S, Y) preserve the reduction in (34)*.*

Proof. Each target realization t leaves the sign B symmetric, so averaging over it gives $\Pr(Y = y \mid T = t) = 1/2$, from which the posterior distribution of T after observing Y equals its prior. Independence also gives the joint likelihood

$$\begin{aligned} \Pr(S = s, Y = y \mid T = t) &= \sum_{b_0 \in \{-1,1\}} \frac{1}{2} \frac{1 + r_B s b_0}{2} \frac{1 + r_A y b_0 t}{2} \\ &= \frac{1}{4}(1 + r_B r_A syt). \end{aligned} \tag{35}$$

With the symmetric prior for T, Bayes' rule gives $\Pr(T = t \mid S = s, Y = y) = (1 + r_B r_A syt)/2$, whose mean is the first expression in (34). Because $T^2 = 1$, the conditional variance is $1 - r_B^2 r_A^2$ for every signal realization, and its expectation gives the stated Bayes risk.

Knowing B allows a person to use the transformed signal $BY = T\epsilon$, whose posterior mean is $r_A BY$ and conditional variance is $1 - r_A^2$. Averaging the risks with and without a foundational observation over its independent receipt event yields $p(1 - r_B^2 r_A^2) + (1 - p) = 1 - pr_B^2 r_A^2$. If a replacement observation depends only on an independent foundation and its own independent noise, it is independent of (T, Y), so it leaves the posterior after Y unchanged. Equality of the information received from professional work and from customer materials gives equality of their conditional distributions and risks. □

Direct transmission of $Z = T\epsilon$ gives posterior mean $r_A Z$ and risk $1 - r_A^2$ even when the person has never observed B, which identifies the feature of the original experiment responsible for the complementarity. With two customer environments indexed by $i \in \{0, 1\}$, the change in acquired competence measured by risk reduction is exactly

$$p_1 (r_B^1)^2 (r_A^1)^2 - p_0 (r_B^0)^2 (r_A^0)^2, \tag{36}$$

so improved explanation quality and reduced access to diagnostic foundations enter the comparison through their actual joint information content.

Corollary A.2 (Transfer across task directions)**.** *Joint uncertainty about several principles has Gaussian distribution $\theta \sim N(\bar{\theta}, \Sigma)$, and a processed exercise supplies $Z = u^{\mathsf{T}}\theta + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$ is independent and $\sigma^2 > 0$. When the subsequent unaided target is $v^{\mathsf{T}}\theta$, the reduction in its minimum expected squared error is*

$$\mathcal{R}_{\text{before}} - \mathcal{R}_{\text{after}} = \frac{(v^{\mathsf{T}}\Sigma u)^2}{\sigma^2 + u^{\mathsf{T}}\Sigma u}. \tag{37}$$

In particular, an exercise has zero transfer exactly when its prior covariance overlap with the target is zero.

Proof. Joint Gaussianity of (θ, Z) gives the posterior covariance below, obtained either by completing the square in the conditional density or by projecting $\theta - \bar{\theta}$ on $Z - u^{\mathsf{T}}\bar{\theta}$,

$$\Sigma_{\text{after}} = \Sigma - \frac{\Sigma u u^{\mathsf{T}}\Sigma}{\sigma^2 + u^{\mathsf{T}}\Sigma u}. \tag{38}$$

Multiplying the covariance reduction on the left by v^{T} and on the right by v gives (37), whose positive denominator establishes the zero-transfer condition. □

A.2 Qualification

Our occupational economy specializes the binary experiment to a single qualification margin with $r_A = r_B = 1$, where qualification requires correct unaided use of the target principle and an eligible novice becomes qualified after processing the revealing pair and passing that assessment. Access to this processed information is supplied by diagnostic cases or purchased instruction, with their resource requirements specified below. If two possible foundational rules predict verified feedback $f_0(z)$ and $f_1(z)$ on a case of type z, their diagnostic set is

$$\mathcal{D} = \{z \mid f_0(z) \neq f_1(z)\}, \qquad M(c) = \int_{\mathcal{R}(c)\cap\mathcal{D}} n(z, c)\, \mathrm{d}z, \tag{39}$$

where n is the purchased case flow and $\mathcal{R}(c)$ contains the cases whose informative reconstruction remains with eligible novices. For fixed arrivals and nested reconstruction sets, removing a positive measure of diagnostic cases strictly reduces M, whereas changes confined to the complement of $\mathcal{D}$ leave it unchanged. Customer-provided diagnostic material contributes to $\mathcal{R}(c)$ whenever the recipient processes the same information, so the interpretation of M follows the assessment and the received feedback.

B Careers

B.1 Ownership and payments

We normalize the population to one and give each person a private qualification state, with $E_t \in (0, 1)$ denoting the qualified share and $1 - E_t$ the novice share. Each qualified person supplies one expert clock hour per unit of time, while a novice supplies one basic hour or produces $b > 0$ units of the numeraire in an outside activity. Qualification travels with its owner across employers, and obsolescence returns the owner to novice status at rate $\delta > 0$. For this qualification margin, the value difference $\Delta_t = V_{E,t} - V_{N,t}$ is an individual asset value measured in units of the numeraire.

Customers whose matters z arrive under measure μ_Z choose a mode $d \in \mathcal{D}(z, c)$, including discovery, professional purchase with different preparation choices, self-service, and an outside option. We denote by $U_d(z, c)$ its benefit after customer time, inference, and other nonlabor resource costs, by j_d and e_d its basic and expert clock hours, and by $m_d \in [0, 1]$ its expected verified transition of an eligible novice into qualification. Modes with zero professional labor may still transmit informative material when an eligible learner actually receives and processes it, with the required contact and processing resources included in the specified mode.

Finite case flows and finite mode requirements bound these opportunities, while each realized transition is assigned to a person who is still a novice at the time of assignment.

Randomized mode shares $\pi_d(z,t)$ summing to one generate basic hours, expert service hours, and diagnostic transition flow according to

$$\begin{aligned} J_t &= \int \sum_d \pi_d(z,t)\, j_d(z,c_t)\, \mathrm{d}\mu_Z(z), \\ D_{S,t} &= \int \sum_d \pi_d(z,t)\, e_d(z,c_t)\, \mathrm{d}\mu_Z(z), \\ M_t &= \int \sum_d \pi_d(z,t)\, m_d(z,c_t)\, \mathrm{d}\mu_Z(z). \end{aligned} \tag{40}$$

Feasibility requires $J_t \le 1 - E_t$ and $D_{S,t} + L_{R,t} \le E_t$, where L_R denotes expert hours purchased by research whenever that sector is present. Throughout the basic-wage benchmark, $J_t < 1 - E_t$, which allows the remaining novices to work in the outside activity and supports their opportunity wage b.

When successful practice rights can be traded, a risk-neutral novice values one unit of transition probability at Δ_t, so competition among eligible purchasers gives a credit $\Delta_t m_d$ for the learning opportunity attached to mode d. Under competitive service provision, this payment finances the fee reduction

$$p_d = b j_d + w e_d - \Delta m_d, \qquad \pi_d > 0 \implies d \in \operatorname*{arg\,max}_{d' \in \mathcal{D}(z,c)} \{U_{d'} - b j_{d'} - w e_{d'} + \Delta m_{d'}\}. \tag{41}$$

For example, the novice may receive gross basic compensation $b j_d$ and pay Δm_d for the placement, or receive their difference as cash compensation, while the provider receives the customer fee and the learning payment and pays $b j_d + w e_d$ for labor. These budgets balance for each mode, including a placement for which the net payment runs from the novice to the provider. Borrowing against future earnings and transferable payments permit the novice to finance the acquisition, while repayment remains the obligation of that individual after a change of employer.

B.2 Private values

Because the final term in the customer's maximand depends on the private asset value, general mode selection determines $D_S(w,c,\Delta)$ and $M(w,c,\Delta)$. For diagnostic transitions assigned symmetrically among novices, the per-person arrival rate is $g = M/(1-E)$, and a price a per successful transition generates the expected net capital gain $g(\Delta - a)$. Formal study supplies a separate qualification hazard $x \ge 0$ at resource cost $\kappa x^2/2$, with $\kappa > 0$, which the worker pays to an independently supplied teaching sector, with instruction using resources charged in the numeraire and consuming no additional professional clock hour from the time endowments in (40).

Individual optimization along differentiable value paths with discount rate $\rho > 0$ gives

$$\begin{aligned} \rho V_N &= b + \max_{x \ge 0} \left\{ -\frac{\kappa x^2}{2} + x\Delta \right\} + g(\Delta - a) + \dot{V}_N, \\ \rho V_E &= w - \delta\Delta + \dot{V}_E. \end{aligned} \tag{42}$$

With complete pricing $a = \Delta$ and a positive skill value, the strictly concave study objective gives $x = \Delta/\kappa$, and subtracting the two Bellman equations yields the coupled career equations

$$\begin{aligned} \dot{\Delta} &= (\rho + \delta)\Delta - (w - b) + \frac{\Delta^2}{2\kappa}, \\ \dot{E} &= M + (1 - E)\frac{\Delta}{\kappa} - \delta E. \end{aligned} \tag{43}$$

By adding diagnostic and formal transitions and subtracting obsolescence, the second equation records the evolution of the qualification share, where independent infinitesimal hazards and assignment to the current novice population prevent duplicate transitions. In expected current cash terms, a novice receives $b-g\Delta-\kappa x^2/2$, while the qualification gains and future earnings enter the value equations, so current compensation and lifetime skill value are distinct observable objects.

B.3 Customer paths

Customer technology paths c_t^ϑ form a differentiable family along which the competitive service-and-training equilibrium joins (41) and (43) with

$$\begin{aligned} D_S(w_t^\vartheta, c_t^\vartheta, \Delta_t^\vartheta) &= E_t^\vartheta, \\ \dot{E}_t^\vartheta &= M(w_t^\vartheta, c_t^\vartheta, \Delta_t^\vartheta) + (1-E_t^\vartheta)\Delta_t^\vartheta/\kappa - \delta E_t^\vartheta, \\ \dot{\Delta}_t^\vartheta &= R\Delta_t^\vartheta - (w_t^\vartheta - b) + (\Delta_t^\vartheta)^2/(2\kappa), \qquad R = \rho + \delta, \end{aligned} \tag{44}$$

with common initial share E_0 and the individual value boundary condition at infinity. On a regular branch with $D_{S,w} < 0$, differentiation is exact to first order, and the sensitivities $y = \partial_\vartheta E$, $z = \partial_\vartheta \Delta$, $h = \partial_\vartheta c$, and $\omega = \partial_\vartheta w$ obey

$$\begin{aligned} \omega &= \frac{y - D_{S,c}h - D_{S,\Delta}z}{D_{S,w}}, \\ \dot{y} &= M_w\omega + M_c h + M_\Delta z + \frac{1-E}{\kappa}z - (x+\delta)y, \\ \dot{z} &= (R+x)z - \omega, \qquad y_0 = 0. \end{aligned} \tag{45}$$

These equations follow by differentiating market clearing and the two career equations, with derivatives evaluated along the benchmark and with the value sensitivity satisfying the differentiated terminal condition whenever the family admits that differentiation. Customer choices change both purchased expert hours and diagnostic opportunities, while the induced wage and qualification values return to the same choices through their labor cost and practice credit.

Research procurement at a strict human corner purchases $L_R = B_R/(w+q\xi)$ expert hours, so holding its budget and input prices fixed gives the wage sensitivity and research labor response

$$\omega = \frac{y - D_{S,c}h - D_{S,\Delta}z}{D_{S,w} - B_R/(w+q\xi)^2}, \qquad \partial_\vartheta L_R = -\frac{B_R}{(w+q\xi)^2}\omega. \tag{46}$$

At a strict machine corner, $L_R = 0$ remains fixed and the first line of (45) applies directly. At an interior mixed allocation, the equal-cost wage is $w = \overline{w}(A)$, so writing $\chi = \partial_\vartheta A$ and $\ell_R = \partial_\vartheta L_R$ gives instead

$$\begin{aligned} \omega &= \overline{w}_A \chi, \\ \ell_R &= y - D_{S,w}\omega - D_{S,c}h - D_{S,\Delta}z, \\ \partial_\vartheta Q_M &= -\frac{L_R\omega + (w+q\xi)\ell_R}{q}. \end{aligned} \tag{47}$$

These expressions follow by differentiating equal packet costs, expert clearing, and the research budget, respectively, and replace the wage-clearing line of (45) while preserving its two career equations. At matched frontier productivity, $\chi = 0$ fixes the mixed wage, while changes in the expert population and customer service demand alter research labor and machine expenditure through the residual equations. For the specified customer interface $c(A) = \kappa_c \log(A/A_0)$, a frontier change contributes $\kappa_c\chi/A$ to h, alongside any direct intervention in customer use. At a boundary between research regimes, the corresponding one-sided formulas follow the regime entered by the perturbation.

C Practice

C.1 Stationary entry

We construct the global comparison by holding the frontier and the elastic demand for nondiagnostic expert services fixed and using a purchased diagnostic portfolio whose selected modes maintain strict margins over the relevant prices and skill values. Variation in the portfolio's diagnostic content then changes the flow M while preserving its expert requirements, and the elastic nondiagnostic portfolio supplies the remaining expert demand. Equilibrium in this portfolio economy gives a continuous inverse wage schedule $w(E)$ independent of M and Δ over the comparison, while the general economy retains the dependence in (41). If research is present, the same statement applies to an inverse wage schedule produced by service and research clearing at the fixed frontier, including a locally flat wage when the two research methods operate together.

Diagnostic matters in a concrete portfolio arrive at rate $n_D > 0$ and retain an expert requirement e_D per purchased case, while their selected customer mode has basic requirement $j_D(c)$ and transition probability $m_D(c)$. Over the compared price range, its customer surplus exceeds each alternative by a common positive margin, so every diagnostic matter is purchased in that mode and gives $M(c) = n_D m_D(c)$. If nondiagnostic expert demand is $D_0(w)$, total demand is $n_D e_D + D_0(w)$ and basic demand is $n_D j_D(c)$, which can satisfy $n_D j_D(c) < 1 - E_+$ uniformly. Customer improvements that reduce $m_D(c)$ change the diagnostic flow without moving this inverse expert-demand curve, while their resource and information consequences remain attached to an explicitly selected mode.

We use an interval $\mathcal{E} = [E_-, E_+] \subset (0,1)$ and a practice interval $\mathcal{M} = [M_-, M_+] \subset [0,\infty)$ on which the specified portfolios are feasible, their basic requirements satisfy $J(E,M) < 1-E$, and w is continuous and nonincreasing with $w(E) > b$. Writing $R = \rho + \delta$, the positive stationary solution of the first equation in (43) gives

$$x(E) = \sqrt{R^2 + \frac{2[w(E) - b]}{\kappa}} - R, \qquad \Delta(E) = \kappa x(E), \qquad F(E,M) = M + (1-E)x(E) - \delta E. \tag{48}$$

We require the following endpoint inequalities to ensure existence throughout the comparison,

$$F(E_-, M_-) > 0, \qquad F(E_+, M_+) < 0, \qquad E_+ < 1. \tag{49}$$

These conditions keep the stationary solution inside a region with a positive novice pool and feasible purchased diagnostic work.

C.2 Replacement

Proposition C.1 (Practice loss and private replacement)**.** *For every $M \in \mathcal{M}$ under* (48) *and* (49)*, there is exactly one stationary expert share $E(M) \in (E_-, E_+)$. If $M^1 < M^0$, its stationary outcomes satisfy*

$$E^1 < E^0, \qquad x^1 \geq x^0, \qquad w^1 \geq w^0, \qquad \Delta^1 \geq \Delta^0, \tag{50}$$

while total formal transitions $T^i = (1-E^i)x^i$ and formal resource spending $C^i = \kappa(1-E^i)(x^i)^2/2$ satisfy

$$0 < T^1 - T^0 < M^0 - M^1, \qquad C^1 > C^0. \tag{51}$$

Whenever $w(E^1) > w(E^0)$, the three weak price and intensity comparisons in (50) *are strict, whereas a constant wage over $[E^1, E^0]$ leaves w, x, and Δ unchanged.*

Proof. Because $w(E) > b$ and w is nonincreasing, $x(E)$ is positive, continuous, and nonincreasing. For $E_2 > E_1$, positivity and monotonicity imply $(1-E_2)x(E_2) < (1-E_1)x(E_1)$, from which $F(E_2, M) < F(E_1, M)$ follows. With the endpoint conditions giving $F(E_-, M) > 0$ and $F(E_+, M) < 0$ for every $M \in \mathcal{M}$, continuity and strict decrease give exactly one root.

At the root for M^0, reducing practice to M^1 gives $F(E^0, M^1) = M^1 - M^0 < 0$, which places the new root strictly below E^0. Monotonicity of w and the strictly increasing relation between w, x, and Δ in (48) give the price comparisons. Moreover, $1 - E^1 > 1 - E^0$ and $x^1 \geq x^0 > 0$ give $T^1 > T^0$. Subtracting the two stationary flow equations gives

$$T^1 - T^0 = (M^0 - M^1) - \delta(E^0 - E^1), \tag{52}$$

whose second term is strictly positive, proving the upper bound in (51). Finally, $C^i = \kappa x^i T^i/2$, so a strict increase in T together with a positive weakly increasing x gives $C^1 > C^0$. □

Differentiability of the wage schedule also gives exact local magnitudes from the preceding proof. With

$$x_E = \frac{w_E}{\kappa(R+x)}, \qquad \Lambda = \delta + x - (1-E)x_E > \delta, \tag{53}$$

implicit differentiation of $F(E(M), M) = 0$ gives

$$\frac{\mathrm{d}E}{\mathrm{d}M} = \frac{1}{\Lambda}, \qquad \frac{\mathrm{d}x}{\mathrm{d}M} = \frac{x_E}{\Lambda}, \qquad \frac{\mathrm{d}w}{\mathrm{d}M} = \frac{w_E}{\Lambda}, \qquad \frac{\mathrm{d}T}{\mathrm{d}M} = \frac{\delta}{\Lambda} - 1 \in (-1, 0). \tag{54}$$

At a locally flat wage, $x_E = 0$ and $\Lambda = \delta + x$, so the formal replacement fraction is $x/(\delta + x)$ even though individual study intensity remains constant. Because its proof uses monotonicity directly, the finite comparison in Proposition C.1 also covers continuous wage schedules with corners.

Customer changes that shift the inverse wage enter the benchmark functions $w(E, c)$ and $M(c)$, under the same differentiability and feasibility conditions. Differentiation at fixed E gives $x_c = w_c/[\kappa(R+x)]$, from which

$$\frac{\mathrm{d}E}{\mathrm{d}c} = \frac{M_c + (1-E)w_c/[\kappa(R+x)]}{\Lambda}. \tag{55}$$

When $w_c \geq 0$, the inequality

$$-M_c > \frac{(1-E)w_c}{\kappa(R+x)} \tag{56}$$

is necessary and sufficient for a declining expert share. Under this inequality, the total wage change $w_c + w_E\,\mathrm{d}E/\mathrm{d}c$ is nonnegative and is positive whenever either term is positive, while $\mathrm{d}T/\mathrm{d}c = -x\,\mathrm{d}E/\mathrm{d}c + (1-E)\,\mathrm{d}x/\mathrm{d}c > 0$. Differentiating the stationary identity $T = \delta E - M$ then gives $0 < (\mathrm{d}T/\mathrm{d}c)/(-M_c) < 1$, which states the extent of private replacement when discovery and practice change together.

Stationary specialization of (42) gives

$$V_N = \frac{b}{\rho} + \frac{\Delta^2}{2\kappa\rho}, \qquad V_E = V_N + \Delta, \qquad \frac{\mathrm{d}(Ew)}{\mathrm{d}M} = \frac{w + Ew_E}{\Lambda}. \tag{57}$$

Thus a strictly increasing skill premium raises both state-contingent lifetime values, while the expert wage bill responds according to $1 + Ew_E/w$, which compares the loss of expert headcount with the increase in earnings per expert. Current novice cash income also contains the practice purchase and teaching payments given above, so its movement follows a separate budget calculation.

C.3 Recipient capacity

Our binary qualification experiment gives a transition after a novice first processes the revealing pair. Repeated copies leave an already qualified recipient's information unchanged. To carry this familiarity effect into a tractable extension, we allow each diagnostic occasion to nominate a person uniformly from the occupational population. Its recipient is novice with probability $1 - E$, while contact with an expert leaves

qualification unchanged. For an informative-contact probability $\ell_d \in [0,1]$ attached to a customer mode, the expected transition and aggregate contact rate are

$$m_d(E) = (1-E)\ell_d, \qquad m(\pi) = \int \sum_d \pi_d \ell_d \, \mathrm{d}\mu_Z, \qquad M = (1-E)m(\pi). \tag{58}$$

Under this single qualification margin, the information gain per offered occasion declines as more recipients already know its principle.

For a novice cohort with independent Poisson contact intensity $m(t)$, temporarily holding formal study and obsolescence at zero, the probability of acquiring the revealed principle by date t is

$$1 - \exp\left[-\int_0^t m(s)\,\mathrm{d}s\right]. \tag{59}$$

Indeed, the probability of receiving zero revealing contacts solves $\dot{p} = -mp$ with $p(0) = 1$, and its complement gives acquisition. As a function of cumulative exposure z, its first derivative is e^{-z} and its second derivative is $-e^{-z}$. Thus diminishing acquisition follows directly from the first revealing observation in the original information experiment.

Each successful transition still belongs to its recipient and has private value Δ. Expected practice revenue is $\Delta(1-E)\ell_d$, so competitive fees and qualification accumulation become

$$F_d = bj_d + we_d - \Delta(1-E)\ell_d, \qquad \dot{E} = (1-E)(m+x) - \delta E. \tag{60}$$

Conditional on being a novice, the contact hazard is m. Complete pricing at $a = \Delta$ removes its net gain from the novice Bellman equation, preserving $x = \Delta/\kappa$ and the original private-value equation. Customer choices now depend directly on E through their expected learning credit.

Proposition C.2 (Replacement with recipient saturation)**.** *Within a feasible strict-margin diagnostic portfolio, varying the contact rate m preserves a continuous nonincreasing inverse wage $w(E) > b$. For the study function $x(E)$ in* (48)*, impose the feasible endpoint inequalities on*

$$(1-E)[m + x(E)] - \delta E. \tag{61}$$

There is then one stationary expert share for each contact rate in the comparison interval. Any finite contact loss $m^1 < m^0$ reduces E, raises formal transitions $T = (1-E)x(E)$ and formal resource expenditure, and gives

$$0 < T^1 - T^0 < M^0 - M^1, \qquad M^i = (1-E^i)m^i. \tag{62}$$

At differentiable points, the formal replacement fraction relative to realized aggregate practice is

$$-\frac{\mathrm{d}T}{\mathrm{d}M} = \frac{x - (1-E)x_E}{\delta + x - (1-E)x_E} \in (0,1). \tag{63}$$

Proof. Because $x(E)$ is positive and nonincreasing, (61) decreases strictly in E. Its endpoint inequalities give one root. Lowering m decreases its value at the original root, so the new root is smaller. Combining an increased novice share with weakly increased positive study intensity raises both T and formal resource expenditure. Stationary flow balance gives

$$M^0 - M^1 = T^1 - T^0 + \delta(E^0 - E^1) > T^1 - T^0 > 0,$$

proving (62).

For local magnitudes, write $\Lambda_s = \delta + m + x - (1-E)x_E > \delta + m$. Differentiating stationarity and then the two transition flows gives

$$E_m = \frac{1-E}{\Lambda_s}, \qquad -T_m = \frac{[x-(1-E)x_E](1-E)}{\Lambda_s}, \qquad M_m = \frac{[\delta + x - (1-E)x_E](1-E)}{\Lambda_s} > 0.$$

Their ratio proves (63). □

Random contact also changes the social effect of an additional incumbent expert. Holding customer mode shares fixed, $M_E = -m$ because one fewer novice remains eligible for future contact. In the service-and-training resource problem, the planner's current-value costate and optimal study intensity consequently satisfy

$$\dot{\lambda} = (\rho + \delta + m)\lambda - (w - b) + \frac{\lambda^2}{2\kappa}, \qquad x = \lambda/\kappa. \tag{64}$$

To derive the extra term, differentiating $\lambda[(1-E)m + I - \delta E]$ at fixed mode shares and aggregate formal transitions I contributes $-\lambda(m+\delta)$ to the Hamiltonian's state derivative. Its formal cost contributes $-\kappa I^2/[2(1-E)^2]$, and the current resource term contributes $w - b$. Applying the costate identity gives (64). Our complete-contract implementation proposition retains its original targeted-placement technology, whose primitive transition yields are state independent. Under this extension, a separate matching effect is associated with the declining mass of eligible recipients.

D Service welfare

D.1 Resources

Our service-and-training allocation problem holds the customer and provider technology paths fixed and abstracts from research procurement, so every expert hour available in this benchmark can be assigned to current professional services. Case menus and their U_d, j_d, e_d, m_d are independent of the aggregate state E, although they may depend on time and the specified technology paths. With $I = (1-E)x$ denoting total formal transitions, a feasible allocation obeys

$$\begin{aligned} &\dot{E} = M(\pi) + I - \delta E, \qquad E(0) = E_0, \qquad 0 \le E < 1, \qquad I \ge 0, \\ &D_S(\pi) \le E, \qquad J(\pi) \le 1 - E, \qquad \pi_d \ge 0, \qquad \sum_d \pi_d = 1. \end{aligned} \tag{65}$$

Allocations with common initial qualification and finite discounted resources are ranked by the social objective

$$\mathcal{V} = \int_0^\infty e^{-\rho t}\left[b(1-E_t) + \int \sum_d \pi_d (U_d - b j_d)\, \mathrm{d}\mu_Z - \frac{\kappa I_t^2}{2(1-E_t)}\right] \mathrm{d}t. \tag{66}$$

Here $b(1-E)$ values the novice time endowment in its outside use and the term bJ removes the outside output displaced by basic services, while the expert constraint allocates scarce qualified hours. Fees, apprentice payments, wages, and tuition finance transfers among participants, with the teaching sector using resources equal to the displayed formal cost. By summing customer and worker payoffs, provider receipts, and teaching-sector receipts, we recover precisely the resource terms in (66).

D.2 Implementation

Proposition D.1 (Competitive practice pricing and implementation)**.** *For the service-and-training economy in* (65) *and* (66)*, a feasible candidate has $J < 1 - E$, $\Delta > 0$, competitive expert wage $w \ge 0$, complementary expert clearing $w(E - D_S) = 0$, the mode choices in* (41)*, and the private career equations* (43)*. If its value path satisfies*

$e^{-\rho t}\Delta_t \to 0$ and all compared objectives are well defined with finite discounted absolute values, this allocation maximizes (66) *among feasible paths sharing E_0. Its social costate can be chosen as $\lambda = \Delta$.*

If an improvement in customer technology optionally expands the mode set while preserving the previous modes with their benefits and resource requirements, the maximal discounted welfare weakly rises, and any complete-contract competitive allocations satisfying these conditions implement the corresponding welfare values.

Proof. Writing $s = 1 - E > 0$, the formal cost $\kappa I^2/(2s)$ is the perspective of the convex function $\kappa x^2/2$. For increments (h, k) in (I, s), its second directional derivative is $\kappa(h - Ik/s)^2/s \geq 0$, which proves joint convexity directly. Every other objective term is linear in E and the mode shares, and the transition and resource constraints are affine, so the feasible allocation problem is concave.

We support the candidate allocation with a social costate λ and expert multiplier w, giving the augmented current-value Hamiltonian

$$\begin{aligned}\mathcal{H}(E, I, \pi, \lambda, w) = b(1 - E) + \int \sum_d \pi_d(U_d - b j_d)\,\mathrm{d}\mu_Z - \frac{\kappa I^2}{2(1-E)} \\ + \lambda\{M(\pi) + I - \delta E\} + w\{E - D_S(\pi)\}.\end{aligned} \tag{67}$$

With $\lambda = \Delta$, its maximization over mode shares reproduces (41), and its derivative in the nonnegative training control gives $\kappa I/(1 - E) = \lambda$, so the candidate interior control is $I = (1 - E)\Delta/\kappa$. Differentiating with respect to the state at fixed controls yields

$$\mathcal{H}_E = w - b - \frac{\kappa I^2}{2(1-E)^2} - \delta\lambda, \qquad \dot{\lambda} = (\rho + \delta)\lambda - (w - b) + \frac{\lambda^2}{2\kappa}. \tag{68}$$

Thus the private equation supplies the social costate equation, while slack novice time has multiplier zero.

Every feasible comparison $(\widetilde{E}, \widetilde{I}, \widetilde{\pi})$ can be evaluated by writing $z = \widetilde{E} - E$ and denoting the objective integrand in square brackets in (66) by G. Concavity and the maximizing mode and training conditions give the supporting inequality

$$\widetilde{G} - G + \lambda\dot{z} \leq (\rho\lambda - \dot{\lambda})z. \tag{69}$$

Complementary clearing removes the candidate expert multiplier term from this inequality, whereas the comparison's term $w(\widetilde{E} - D_S(\widetilde{\pi}))$ is nonnegative by feasibility, and removing that term preserves the displayed inequality. Even when the comparison exhausts its novice time, the candidate's zero basic-time multiplier provides the supporting inequality over the full feasible comparison set.

Multiplication by $e^{-\rho t}$ and integration through T yields

$$\int_0^T e^{-\rho t}(\widetilde{G}_t - G_t)\,\mathrm{d}t \leq -e^{-\rho T}\lambda_T z_T, \tag{70}$$

since $z_0 = 0$. Both expert shares lie in $[0, 1)$, so $|z_T| \leq 1$ and the stated discounted value condition sends the right-hand side to zero. Convergence of the discounted objectives proves global optimality.

Retaining every old mode under the expanded customer technology preserves every old feasible control and state path. Maximization over the expanded set gives a value at least as large as its old value, and the verification result identifies each qualifying competitive equilibrium with its corresponding optimum. Whenever a feasible new path has value strictly above the old optimum, the welfare increase is strict. □

Because Proposition D.1 compares complete feasible allocations, the welfare gain from an optional technology can coexist with a reduced expert share whenever the improved service options save resources sufficient to cover the value of displaced learning. Its price support belongs to the competitive service-and-training benchmark, whose mode prices value all current resources and private qualification gains. Research procurement and provider markups have the separately specified objectives and wedges developed in their respective parts of the model.

E Contract limits

E.1 Learning payments

At a complete-contract benchmark, a negligible class of current cases allows a direct comparison at the common shadow value $\lambda = \Delta > 0$. For each case in this class, the unprepared mode uses one basic hour and gives a diagnostic transition probability $m > 0$, whereas customer preparation replaces the basic hour at nonnegative resource cost r_C and removes this transition, with the service benefit and expert requirement held fixed. Under a restriction on the apprenticeship contract, the fee receives a learning credit of only am, where $0 \le a < \Delta$, and applying the restriction to this negligible class leaves the benchmark prices and continuation values fixed to first order.

E.2 Preparation

Proposition E.1 (A binding compensation floor)**.** *Under the preceding local comparison, customer preparation is strictly preferred when $b - r_C > am$, and its marginal contribution to discounted resource welfare has the sign of $b - r_C - \Delta m$. Every saving in the interval*

$$am < b - r_C < \Delta m \tag{71}$$

produces privately chosen preparation with a strictly negative marginal welfare effect.

For a positive cash compensation floor $s \in (0, b)$ that also prohibits separate tuition charges by the placement provider, a binding restriction has $am = b - s < \Delta m$, and the interval is nonempty for a nonnegative customer cost exactly when $\max\{0, b - \Delta m\} < s$. For example, the symbolic values

$$b = 1, \qquad \Delta m = \frac{3}{4}, \qquad s = \frac{1}{2}, \qquad r_C = \frac{3}{8} \tag{72}$$

give $am = 1/2 < b - r_C = 5/8 < \Delta m = 3/4$.

Proof. With a permitted learning credit am, the unprepared mode's basic component of the fee is $b - am$, while preparing the case replaces that fee component by r_C. Customer preparation consequently gains $b - r_C - am$, whose positivity gives the private choice. In the benchmark planner's Hamiltonian, the same change saves the current resource amount $b - r_C$ and removes a transition valued at $\lambda m = \Delta m$, giving the social expression and (71). At a date t, a marginal mass of such cases scales this expression by the positive discount factor $e^{-\rho t}$, so its sign also gives the marginal discounted welfare sign.

Net cash compensation $b - am$ for the placement must remain at least s under the stated rule, and the prohibition on separate tuition prevents an offsetting charge outside that compensation. Competition passes through $am = \min\{\Delta m, b - s\}$, whose binding value is $b - s$. Substituting it into (71) gives

$$b - \Delta m < r_C < s, \qquad r_C \ge 0. \tag{73}$$

Existence of a nonnegative r_C satisfying both strict inequalities is equivalent to $s > \max\{0, b - \Delta m\}$, and direct substitution verifies (72). If the floor equals zero, a binding restriction gives $am = b$, while $b - r_C \le b$ for every nonnegative r_C, so the strict private-preparation inequality has an empty feasible set. With an unbound floor, full pricing gives $a = \Delta$ and the two thresholds coincide. □

Applying the restriction to a positive case mass changes the equilibrium allocation, whose private value Δ and social costate λ are distinct endogenous objects. Their corresponding local private and social comparisons are $b - r_C - am$ and $b - r_C - \lambda m$, and their difference is $(\lambda - a)m$, so the benchmark equality used in Proposition E.1 has a precise role in the marginal experiment.

F Service quality

F.1 Fees and losses

Our nondiagnostic portfolio has a fixed finite franchised provider set, with each provider i assigned a separate customer population whose isoelastic demand is $Q_i = N_i(c)P_i^{-\varepsilon}$, where $\varepsilon > 1$ is common across providers and the demand of each population is independent of all other providers' prices. Their common service and risk technologies imply the same optimal validation and generalized price at a given wage, so adding their disjoint customer populations gives aggregate demand $Q = N(c)P^{-\varepsilon}$ with $N(c) = \sum_i N_i(c)$. Each completed case uses $e(c) > 0$ expert hours for generation, $v \geq 0$ expert hours for observable validation, and $k(c) \geq 0$ units of other resources, while an unresolved error causes monetary loss $L > 0$ with probability $z(c, v) = \pi(c)e^{-v}$ for $0 \leq \pi(c) \leq 1$. Consumers with constant absolute risk aversion $\gamma \geq 0$ assign this binary loss the certainty-equivalent cost

$$r_\gamma(z) = \begin{cases} \gamma^{-1}\log\{1 + z(e^{\gamma L} - 1)\}, & \gamma > 0, \\ Lz, & \gamma = 0. \end{cases} \tag{74}$$

Charging a money fee f per completed case gives generalized price $P = f + r_\gamma(z)$ and demand $Q = N(c)P^{-\varepsilon}$, so a franchised provider chooses its fee and validation to maximize

$$\{f - w[e(c) + v] - k(c)\}N(c)\{f + r_\gamma(z(c, v))\}^{-\varepsilon}. \tag{75}$$

These fees are fixed per completed case at the given market prices, and their equilibrium level can change when the wage or customer environment changes.

F.2 Validation

Lemma F.1 (Optimal validation and decreasing expert demand)**.** *For $w > 0$ in the preceding economy, we define*

$$\mathcal{T}(w, c) = \min_{v \geq 0}\{w[e(c) + v] + k(c) + r_\gamma(\pi(c)e^{-v})\}, \qquad \mathfrak{m}_f = \frac{\varepsilon}{\varepsilon - 1}. \tag{76}$$

Optimal fees satisfy $P = \mathfrak{m}_f\mathcal{T}$, and optimal validation satisfies $w + \partial_v r_\gamma = 0$ whenever $v > 0$, with $w + \partial_v r_\gamma \geq 0$ at $v = 0$. For risk-neutral customers,

$$v^*(w, c) = \left[\log\frac{L\pi(c)}{w}\right]_+ \tag{77}$$

when $\pi(c) > 0$, with $v^* = 0$ when $\pi(c) = 0$. Expert demand is

$$D_S(w, c) = \mathcal{T}_w(w, c)N(c)\{\mathfrak{m}_f\mathcal{T}(w, c)\}^{-\varepsilon}, \tag{78}$$

which is strictly decreasing in w wherever its derivative exists, and continuously decreasing across an entry or exit of validation.

Proof. Expected exponential utility at $\gamma > 0$ and initial wealth y is $-e^{-\gamma y}\{1 + z(e^{\gamma L} - 1)\}$ for the binary loss, so equating this expression to $-e^{-\gamma(y-r)}$ gives (74), whose limit as $\gamma \to 0$ is Lz. For a fixed validation choice v, profit can be expressed as $N(P - T_v)P^{-\varepsilon}$, where $T_v = w(e + v) + k + r_\gamma(z) > 0$. Its derivative has the sign of $\varepsilon T_v - (\varepsilon - 1)P$, giving the global maximizer $P = \mathfrak{m}_f T_v$ and a maximized profit proportional to $T_v^{1-\varepsilon}$. Thus the provider minimizes T_v over validation.

Validation-dependent risk at $\gamma > 0$ can be written using $K = \pi(c)(e^{\gamma L} - 1)$ as $\gamma^{-1}\log(1 + Ke^{-v})$, whose second derivative is $Ke^{-v}/[\gamma(1 + Ke^{-v})^2] \geq 0$. For $\gamma = 0$, its second derivative is $L\pi(c)e^{-v} \geq 0$. Since wv grows without bound and $e > 0$, the minimizer exists, is unique whenever $\pi(c) > 0$, and obeys the stated

boundary and interior conditions, while $\pi(c) = 0$ gives the unique minimizer zero. In the risk-neutral interior, the first-order condition is $w = L\pi(c)e^{-v}$, which gives (77) together with the boundary condition.

Each fixed validation choice v gives an affine function T_v of w, so its minimized value $\mathcal{T}$ is concave in w. Uniqueness and the envelope theorem give $\mathcal{T}_w = e + v^* > 0$, from which (78) follows by multiplying hours per case by purchased quantity. At differentiable points of the demand curve,

$$\partial_w D_S = N\mathfrak{m}_f^{-\varepsilon}\mathcal{T}^{-\varepsilon}\left\{\mathcal{T}_{ww} - \varepsilon\frac{\mathcal{T}_w^2}{\mathcal{T}}\right\} < 0, \tag{79}$$

because concavity gives $\mathcal{T}_{ww} \leq 0$. Across a validation boundary, continuity of the unique optimizer gives continuous hours, while $e + v^*$ is nonincreasing and $\mathcal{T}$ is strictly increasing, so the product in (78) remains strictly decreasing. □

F.3 Service surplus

Inverse generalized demand at fixed c is $\mathcal{P}(Q, c) = \{N(c)/Q\}^{1/\varepsilon}$, with integral $\mathcal{B}(Q, c) = \int_0^Q \mathcal{P}(y, c)\,\mathrm{d}y$, which is finite because $\varepsilon > 1$. Before valuing the expert time endowment in another occupation, aggregate certainty-equivalent service surplus is

$$\mathcal{S}(E_S) = \mathcal{B}(Q, c) - k(c)Q - r_\gamma(z(c, v))Q, \qquad E_S = Q[e(c) + v]. \tag{80}$$

At an interior validation choice, differentiation gives

$$\begin{aligned}\mathrm{d}\mathcal{S} &= (P - k - r_\gamma)\,\mathrm{d}Q - Q\partial_v r_\gamma\,\mathrm{d}v \\ &= w\,\mathrm{d}E_S + (\mathfrak{m}_f - 1)\mathcal{T}\,\mathrm{d}Q,\end{aligned} \tag{81}$$

where the validation condition and $P = \mathfrak{m}_f\mathcal{T}$ supply the second equality. On an interval with $v = 0$, the same identity follows from $\mathrm{d}v = 0$. Since both purchased quantity and expert demand decline with w, their ratio of wage derivatives gives $Q_{E_S} > 0$ at regular points, and hence

$$\frac{\mathrm{d}\mathcal{S}}{\mathrm{d}E_S} = w + (\mathfrak{m}_f - 1)\mathcal{T}Q_{E_S} > w. \tag{82}$$

If these additional expert hours arise by transferring workers from novice outside production, their current resource contribution is smaller by b per hour, so the associated current qualification return is $w - b + (\mathfrak{m}_f - 1)\mathcal{T}Q_{E_S}$. Fee receipts, worker wages, and profits paid to the fixed franchise owners cancel in this aggregate account, while the markup term measures the marginal customer value of additional cases beyond the privately received expert wage. With $\mathfrak{m}_f > 1$, this positive wedge replaces the zero-wedge competitive price support used by Proposition D.1, even when complete apprenticeship contracts credit every learning opportunity at its private value Δm. Provider market power thus enters the distinction between private and social skill values through the conditional current wedge.

G Research

G.1 Methods

Qualified workers supply expertise used both in customer services and in the generation or evaluation of research in the same field, so software and engineering give the shared labor market a direct occupational interpretation. Each member of the expert population supplies one clock hour, and the fraction $E \in (0, 1)$ measures the available quantity of that rival input. With customer capability c, private qualification value Δ, and hourly wage w, customer choices generate expert demand $D_S(w, c, \Delta)$ and diagnostic qualification flow $M(w, c, \Delta)$.

Funding comes from a public research fund that receives a fixed flow $B_R > 0$ from lump-sum contributions by an independently endowed numeraire sector, whose endowment covers those contributions. Its mandate purchases the greatest flow of completed validated research packets at competitive input prices. With computation priced at its opportunity cost $q > 0$, human research hours also require $\xi \geq 0$ computation units per hour, while the independent teaching sector continues to supply the formal training purchased by novices.

At positive research capability r, completed human packets require generation hours $r^{-\alpha}$ and validation hours $v_R r^{-\nu}$, where $v_R > 0$ and $\alpha, \nu \geq 0$. Completed autonomous packets require generation computation $r^{-\mu}$ and machine-validation computation $m_R r^{-\zeta}$, where $m_R > 0$ and $\mu, \zeta \geq 0$. With mutually distinct targets assigned to the two methods, their requirements and costs are

$$a_H(r) = r^{-\alpha} + v_R r^{-\nu}, \qquad k_H(r, w) = a_H(r)(w + q\xi), \tag{83}$$

$$a_M(r) = r^{-\mu} + m_R r^{-\zeta}, \qquad k_M(r) = q a_M(r). \tag{84}$$

Completion requires both stages of the chosen method. For a human packet flow z_H, feasible generation and validation inputs satisfy

$$L_G \geq r^{-\alpha} z_H, \qquad L_V \geq v_R r^{-\nu} z_H, \qquad L_R = L_G + L_V,$$

so positive input prices give $L_R = a_H(r) z_H$. Human validation capacity equals $r^{\nu} L_V / v_R$, with its quantity determined by purchased expert time. Corresponding machine constraints give $Q_M = a_M(r) z_M$, including the computation used to validate and integrate autonomous output. Complete autonomous production becomes available in the specified region $A \geq A_M$, where feasibility includes every stage represented by its packet.

With $A \geq A_0 > 0$ measuring cardinal frontier productivity, the main specification sets $r = A$ and

$$c(A) = \kappa_c \log(A/A_0), \qquad \dot{A} = A^{-\phi}(z_H + z_M), \qquad \kappa_c > 0, \quad \phi > 0. \tag{85}$$

Holding packet output fixed, the factor $A^{-\phi}$ raises the difficulty of obtaining another absolute frontier increment. In proportional-growth units, the corresponding difficulty exponent is $\phi + 1$. Customer-use comparisons can vary the interface $c_\vartheta(A)$, with $c_0(A) = c(A)$, while keeping the research requirement functions a_H, a_M common, so dc at a fixed frontier records a change in that interface. Research and customer services share the market and budget identities

$$E = D_S(w, c, \Delta) + L_R, \qquad (w + q\xi) L_R + q Q_M = B_R. \tag{86}$$

G.2 Allocation

Proposition G.1 (Competitive research allocation)**.** *Service demand at fixed (A, E, c, Δ) is continuous and strictly decreasing in w on a positive wage interval, and both equations*

$$D_S(w_0, c, \Delta) = E, \qquad D_S(w_1, c, \Delta) + \frac{B_R}{w_1 + q\xi} = E \tag{87}$$

have solutions in that interval. These solutions are unique and satisfy $w_0 < w_1$. With

$$\overline{w}(A) = q \frac{a_M(A)}{a_H(A)} - q\xi, \tag{88}$$

the human-only feasible region has $w = w_1$, $L_R = B_R/(w_1 + q\xi)$, and $Q_M = 0$. Where both methods are available, equilibrium allocations satisfy

$$\begin{aligned} w_1 \leq \overline{w} &\implies (w, L_R, Q_M) = (w_1, B_R/(w_1 + q\xi), 0), \\ w_0 \geq \overline{w} &\implies (w, L_R, Q_M) = (w_0, 0, B_R/q), \\ w_0 < \overline{w} < w_1 &\implies w = \overline{w}, \quad L_R = E - D_S(\overline{w}, c, \Delta), \\ & \qquad Q_M = [B_R - (\overline{w} + q\xi) L_R]/q. \end{aligned} \tag{89}$$

With packet flows $z_H = L_R/a_H(A)$ and $z_M = Q_M/a_M(A)$, the frontier follows

$$F(A, E, c, \Delta) = A^{-\phi}\frac{B_R}{k_*}, \qquad k_* = \min_{j \in \mathcal{F}(A)} k_j, \tag{90}$$

where $\mathcal{F}(A)$ contains the feasible methods and all costs use the cleared wage.

Proof. Strictly decreasing service demand and the strictly decreasing function $B_R/(w + q\xi)$ imply uniqueness of both roots in (87). At w_0, combined human-research and service demand exceeds E, so its root satisfies $w_1 > w_0$.

Given a wage, the fund solves a linear procurement problem with positive packet prices and a binding expenditure constraint, so it purchases only the cheaper feasible method, with both methods purchased at equal costs. Comparing (83) and (84) gives $k_H \le k_M$ exactly when $w \le \overline{w}$. Human-only purchasing requires the wage w_1, giving the first condition, while machine-only purchasing leaves the service wage w_0, giving the second.

When $w_0 < \overline{w} < w_1$, equality of packet prices fixes the mixed wage at $\overline{w}$. Strictly decreasing service demand gives $E - D_S(\overline{w}, c, \Delta) > 0$. Since total demand evaluated at $\overline{w} < w_1$ exceeds E, it also gives

$$E - D_S(\overline{w}, c, \Delta) < \frac{B_R}{\overline{w} + q\xi}.$$

Thus both physical allocations in the third branch of (89) are positive, and the labor market and budget determine them uniquely. At either endpoint the mixed expression agrees with the adjacent corner. Expenditure on every active packet equals the common minimum unit cost, giving total output B_R/k_* and then (90). □

G.3 Expertise and private values

Proposition G.2 (The shared-wage research feedback)**.** *At a strict human corner, differentiable service demand satisfies $D_{S,w} < 0$. For fixed (A, c, Δ), we define*

$$h = w + q\xi, \qquad \mathcal{J} = -D_{S,w} + \frac{B_R}{h^2} > 0. \tag{91}$$

Local headcount effects satisfy

$$w_E = -\frac{1}{\mathcal{J}} < 0, \qquad \frac{F_E}{F} = \frac{1}{h\mathcal{J}} > 0, \qquad (L_R)_E = \frac{B_R}{h^2\mathcal{J}} > 0. \tag{92}$$

Holding A, B_R, q, ξ fixed, a perturbation of headcount and the customer environment has the exact first-order decomposition

$$\mathrm{d}\log F = \frac{\mathrm{d}E - \mathrm{d}D_S|_w}{h\mathcal{J}}, \qquad \mathrm{d}D_S|_w = D_{S,c}\,\mathrm{d}c + D_{S,\Delta}\,\mathrm{d}\Delta + \mathrm{d}D_S^{\mathrm{direct}}|_w. \tag{93}$$

At an interior mixed allocation, fixed A gives $w_E = F_E = 0$, $(L_R)_E = 1$, and $(Q_M)_E = -h/q$. At a strict machine corner, $F_E = 0$ and $w_E = 1/D_{S,w} < 0$.

Proof. Differentiation of human-corner clearing gives

$$\mathrm{d}E = \left(D_{S,w} - \frac{B_R}{h^2}\right)\mathrm{d}w + \mathrm{d}D_S|_w = -\mathcal{J}\,\mathrm{d}w + \mathrm{d}D_S|_w.$$

Holding the customer environment fixed produces $w_E = -1/\mathcal{J}$, and differentiation of $L_R = B_R/h$ produces its stated headcount response. Human frontier output satisfies

$$\log F = \log B_R - \phi\log A - \log a_H(A) - \log h,$$

so $\mathrm{d}\log F = -\mathrm{d}w/h$ at matched A. Substitution proves both (92) and (93). In a mixed allocation, equal costs fix $w = \overline{w}(A)$ and $F = B_R A^{-\phi}/[qa_M(A)]$, while the residual labor and budget identities give the two allocation derivatives. In the machine corner, the same expression fixes F, and differentiation of service-only clearing gives the wage derivative. □

Two human-corner economies compared at a common frontier, with $E^{(1)} \le E^{(0)}$ and $D_S^{(1)}(w) \ge D_S^{(0)}(w)$ at every common wage, satisfy $w^{(1)} \ge w^{(0)}$ and $F^{(1)} \le F^{(0)}$. Indeed, economy 1's total demand at economy 0's wage is at least $E^{(0)}$, which is at least $E^{(1)}$, so strictly decreasing demand requires a weakly higher wage to clear. Strictness in the headcount inequality or the demand inequality at the comparison wage gives a strict output ranking. Extending this finite matched-frontier effect to complete paths requires those orders along the evolving matched states.

Private careers retain opportunity pricing $a = \Delta$, which yields the Bellman equations

$$\rho V_N = b + \max_{x \ge 0}\{-\kappa x^2/2 + x\Delta\} + g(\Delta - a) + \dot{V}_N, \tag{94}$$

$$\rho V_E = w - \delta\Delta + \dot{V}_E, \qquad g = \frac{M(w, c, \Delta)}{1 - E}. \tag{95}$$

On the positive-premium region, $\Delta \ge 0$ gives $x = \Delta/\kappa$, so the joint economy obeys

$$\dot{E} = M(w, c(A), \Delta) + (1 - E)\frac{\Delta}{\kappa} - \delta E, \tag{96}$$

$$\dot{\Delta} = (\rho + \delta)\Delta - (w - b) + \frac{\Delta^2}{2\kappa}, \tag{97}$$

$$\dot{A} = F(A, E, c(A), \Delta). \tag{98}$$

Over a finite horizon T with finite states, specified terminal continuation values $V_N(T) = \overline{V}_N$ and $V_E(T) = \overline{V}_E$ close the private career problem, with $\overline{\Delta} = \overline{V}_E - \overline{V}_N \ge 0$. Their difference supplies the terminal condition $\Delta(T) = \overline{\Delta}$, while inherited headcount and frontier productivity specify $E(0)$ and $A(0)$. Global paths on $[0, \infty)$ use worker value and transversality conditions at infinity to select the forward value. Equations (86)–(90) determine wages and the research method at each date, so future research earnings enter today's privately financed qualification decision over the specified career horizon.

Proposition G.3 (Private qualification value and future wages)**.** *For two nonnegative qualification-value paths satisfying* (97) *on a common finite horizon* $[0, T]$*, with common* $(\rho, \delta, \kappa, b)$ *and specified terminal values, we define* $d(t) = \Delta_1(t) - \Delta_0(t)$ *and*

$$\mathcal{K}(t, s) = \exp\left\{-\int_t^s \left[\rho + \delta + \frac{\Delta_1(u) + \Delta_0(u)}{2\kappa}\right] \mathrm{d}u\right\}. \tag{99}$$

Their difference has the finite-horizon representation

$$d(t) = \mathcal{K}(t, T)d(T) + \int_t^T \mathcal{K}(t, s)[w_1(s) - w_0(s)]\,\mathrm{d}s. \tag{100}$$

For bounded global value paths on $[0, \infty)$*, the corresponding representation is*

$$d(t) = \int_t^\infty \mathcal{K}(t, s)[w_1(s) - w_0(s)]\,\mathrm{d}s. \tag{101}$$

With common terminal qualification values in the finite-horizon comparison, a weakly higher future wage path yields a weakly higher present qualification value and formal-training intensity, with strictness whenever the

future wage difference is positive on a set of positive measure. Bounded global paths satisfy the same wage ordering. At stationary wages above b, the positive value solves

$$(\rho+\delta)\Delta+\frac{\Delta^2}{2\kappa}=w-b,\qquad \frac{\mathrm{d}\Delta}{\mathrm{d}w}=\frac{1}{\rho+\delta+\Delta/\kappa}>0. \tag{102}$$

Proof. Subtracting the two value equations gives

$$\dot{d}(t)=\left[\rho+\delta+\frac{\Delta_1(t)+\Delta_0(t)}{2\kappa}\right]d(t)-[w_1(t)-w_0(t)].$$

Multiplication by the integrating factor and integration from t to T gives (100), whose terminal term vanishes under equal terminal qualification values. For global paths, bounded values and a discount rate at least $\rho+\delta>0$ send that term to zero as $T\to\infty$, proving (101). Strictly positive integral weights establish both wage-path comparisons. Setting $\dot{\Delta}=0$ yields the stationary identity, whose derivative gives the final expression. □

Under the stationary practice benchmark with M and D_S independent of Δ, an interior mixed-research region holds the common wage fixed at $\overline{w}(A)$, so (102) fixes Δ and x. Differentiating $M+(1-E)x-\delta E=0$ then gives

$$E_M=\frac{1}{\delta+x},\qquad w_M=\Delta_M=x_M=0,\qquad \frac{\mathrm{d}[(1-E)x]}{\mathrm{d}M}=-\frac{x}{\delta+x}. \tag{103}$$

With $x>0$, a loss of practice reduces expert headcount and raises total formal transitions by expanding the novice population, while individual study intensity remains constant over this region.

G.4 Income

Proposition G.4 (Factor receipts and automation)**.** *Research labor income, compute-service receipts, and labor's research expenditure share satisfy*

$$W_R=wL_R,\qquad C_R=q(\xi L_R+Q_M),\qquad W_R+C_R=B_R,\qquad s_R^L=\frac{wL_R}{B_R}. \tag{104}$$

In the human corner $s_R^L=w/(w+q\xi)$, and in the machine corner $s_R^L=0$. At a strict human corner, fixed (A,c,Δ) gives

$$\frac{\partial(Ew)}{\partial E}=w-\frac{E}{\mathcal{J}},\qquad \frac{\partial s_R^L}{\partial E}=-\frac{q\xi}{h^2\mathcal{J}}. \tag{105}$$

At fixed (A,E,c,Δ), access to the complete machine method weakly lowers the common wage, the skill premium $w-b$, and total expert wage income Ew, relative to the human-only allocation.

Proof. Substitution of the budget identity in (86) gives (104), and the two corner allocations give their labor shares. Applying the product rule with $w_E=-1/\mathcal{J}$ gives the first derivative in (105), while differentiation of $w/(w+q\xi)$ gives the second. After the machine method becomes available, Proposition G.1 places the wage in $[w_0,w_1]$, where w_1 is the original human-only wage. Multiplication by fixed positive E and subtraction of fixed b establish the income and premium statements. □

Compute receipts belong to the owners of the supplied computation and include the costs of producing that service, with zero excess profit under competitive constant unit cost q. Because total expert income has the derivative $w-E/\mathcal{J}$, higher hourly scarcity earnings can accompany either direction of the total expert wage bill. Common research expenditure B_R permits different physical combinations of expert hours and computation, and s_R^L records the research sector's budget share.

H Equilibrium

H.1 Cleared wages

We write $X = (E, A)$ for the inherited states and retain Δ as the private qualification value. For the results in this appendix, $D(E, w, A, \Delta)$ and $M(E, w, A, \Delta)$ include the customer interface $c(A)$. Their direct dependence on E disappears in the original targeted-placement economy, while the recipient-capacity extension in Appendix C uses that argument. With $R = \rho + \delta$, we define the cleared learning flow and state field by

$$\begin{aligned} \widehat{M}(X, \Delta) &= M(E, W(X, \Delta), A, \Delta), \\ H(X, \Delta) &= \begin{pmatrix} \widehat{M}(X, \Delta) + (1 - E)\Delta/\kappa - \delta E \\ F(X, \Delta) \end{pmatrix}, \\ G(X, \Delta) &= R\Delta - [W(X, \Delta) - b] + \frac{\Delta^2}{2\kappa}. \end{aligned} \tag{106}$$

Our joint equations are $\dot{X} = H(X, \Delta)$ and $\dot{\Delta} = G(X, \Delta)$, with F given by research procurement at the cleared wage.

On a compact region with $0 < E < 1$ and $A > 0$, continuously differentiable demand with $D_w \le -\sigma < 0$ gives regular wage roots whenever both clearing equations have solutions in a common compact interval. We denote those roots by w_0 and w_1, as in Proposition G.1. Once both research methods are feasible, their allocation can be written as

$$W(X, \Delta) = \min\{w_1(X, \Delta), \max\{w_0(X, \Delta), \overline{w}(A)\}\}, \qquad \overline{w} = q\frac{a_M}{a_H} - q\xi. \tag{107}$$

For the human-only feasible region, $W = w_1$. Each root is continuously differentiable by implicit differentiation. Finite maxima and minima preserve local Lipschitz continuity, so (107) remains Lipschitz across the endogenous human, mixed, and machine boundaries. Those boundaries can change derivatives while preserving the wage level.

For $j \in \{0, 1\}$ and $\mathcal{J}_j = -D_w + jB_R/(w_j + q\xi)^2$, the root derivatives are

$$(w_j)_E = -\frac{1 - D_E}{\mathcal{J}_j}, \qquad (w_j)_\Delta = \frac{D_\Delta}{\mathcal{J}_j}, \qquad (w_j)_A = \frac{D_A}{\mathcal{J}_j}. \tag{108}$$

Here each derivative of D holds its wage argument fixed. Positive packet costs on a compact region also keep F Lipschitz when the primitive functions are Lipschitz. These observations provide the price regularity used in the existence argument below.

H.2 Finite careers

For a horizon T, inherited expertise and frontier productivity give $X(0) = X_0$, while a specified continuation occupation or contractual terminal payoff gives $\Delta(T) = \overline{\Delta} \ge 0$. Our terminal condition supplies the future earnings component of today's investment. Proposition D.1 verifies optimality of a qualifying competitive allocation in the service-and-training benchmark. We now establish existence and uniqueness for the joint boundary-value system under explicit local bounds.

Proposition H.1 (Finite-horizon equilibrium)**.** *Within a fixed availability region, we take a compact rectangular state set K_X containing X_0 in its interior and a qualification-value interval $[0, D]$. Throughout $K_X \times [0, D]$, novice time is strictly feasible, the cleared wage satisfies $b \le W \le W_{\max}$, and H, W have joint Lipschitz constants L_H, L_W in the product maximum norm. We choose D with*

$$D \ge \overline{\Delta}, \qquad RD + \frac{D^2}{2\kappa} \ge W_{\max} - b. \tag{109}$$

For the distance $m_X > 0$ from X_0 to the boundary of K_X and a uniform bound $H_{\max}$ on the state field, the inequalities

$$TH_{\max} < m_X, \qquad \max\left\{TL_H, \frac{1-e^{-RT}}{R}L_W\right\} < 1 \tag{110}$$

give exactly one equilibrium in $K_X \times [0, D]$ with $X(0) = X_0$ and $\Delta(T) = \overline{\Delta}$. It includes endogenous crossings between human, mixed, and machine research. For a Lipschitz terminal payoff $\Delta(T) = \Psi(X(T)) \in [0, D]$, the same conclusion follows when the second inequality is replaced by

$$\max\left\{TL_H, L_\Psi TL_H + \frac{1-e^{-RT}}{R}L_W\right\} < 1. \tag{111}$$

Proof. For continuous trial paths $Y = (X, \Delta)$ in the stated region with $X(0) = X_0$, we define the updated state path

$$\widetilde{X}(t) = X_0 + \int_0^t H(X(s), \Delta(s))\,\mathrm{d}s. \tag{112}$$

Its displacement from X_0 is bounded by $TH_{\max} < m_X$, so it stays in K_X. We define the updated qualification value through

$$\dot{\widetilde{\Delta}} = R\widetilde{\Delta} + \frac{\widetilde{\Delta}^2}{2\kappa} - [W(X(t), \Delta(t)) - b], \qquad \widetilde{\Delta}(T) = \overline{\Delta}. \tag{113}$$

After reversing time, the drift at zero is $W - b \geq 0$ and the drift at D is at most $W_{\max} - b - RD - D^2/(2\kappa) \leq 0$. Scalar existence and uniqueness and these inward inequalities give a unique solution in $[0, D]$ throughout the horizon. Thus the two updates map the complete space of feasible continuous trial paths into itself.

For two trial paths, their updated state difference is bounded by TL_H times their original supremum distance. Their updated value difference v satisfies

$$\dot{v} = \left[R + \frac{\widetilde{\Delta}_1 + \widetilde{\Delta}_0}{2\kappa}\right] v - [W(Y_1) - W(Y_0)], \qquad v(T) = 0.$$

Variation of constants and nonnegative qualification values give

$$\|v\|_\infty \leq \frac{1-e^{-RT}}{R} L_W \|Y_1 - Y_0\|_\infty.$$

Under (110), the update is a contraction. Its unique fixed point solves the joint equations and their initial and terminal conditions. Every equilibrium inside the region is a fixed point, which proves uniqueness there.

For a terminal function Ψ, first form $\widetilde{X}$ and then use $\widetilde{\Delta}(T) = \Psi(\widetilde{X}(T))$ in (113). Their updated terminal-value difference is bounded by $L_\Psi TL_H$ times the original path distance. Adding this term to the preceding value bound gives (111). Its fixed point satisfies the required terminal identity.

To recover full worker values, a terminal novice payoff $\overline{V}_N$ and expert payoff $\overline{V}_N + \overline{\Delta}$ give

$$\begin{aligned} V_N(t) &= e^{-\rho(T-t)}\overline{V}_N + \int_t^T e^{-\rho(s-t)}\left[b + \frac{\Delta(s)^2}{2\kappa}\right]\mathrm{d}s, \\ V_E(t) &= V_N(t) + \Delta(t). \end{aligned} \tag{114}$$

Direct differentiation verifies both worker Bellman equations under complete practice pricing. Strict concavity of the study objective gives its unique maximizer $x = \Delta/\kappa$. Integrating the Bellman inequality along the worker's qualification process gives the value bound for every feasible training policy, with equality under this maximizer. Bounded hazards follow from the positive novice share. Customer and research choices clear at the constructed wage, so the fixed point is a rational-expectations equilibrium. Because the proof uses uniform bounds and time integrals, it also applies to bounded measurable time coefficients, including changes at specified calendar dates. □

Within this result, bounded wages limit the return to qualification, novice slack preserves the basic outside activity, and the horizon bound limits feedback from current practice credits into future earnings. Every finite Lipschitz terminal function permits a sufficiently short horizon. Longer horizons can be studied by the local determinacy criterion that follows.

H.3 Determinacy and switches

Proposition H.2 (Local selection of the jump value)**.** *Along a differentiable finite-horizon equilibrium, start the initial-value equations at* (X_0, Δ_0) *and define*

$$S(\Delta_0) = \Delta(T \mid \Delta_0) - \Psi(X(T \mid \Delta_0)). \tag{115}$$

For $u = \partial_{\Delta_0}(X, \Delta)$*, solve*

$$\dot{u} = \begin{pmatrix} H_X & H_\Delta \\ G_X & G_\Delta \end{pmatrix} u, \qquad u(0) = (0, 0, 1)^\mathsf{T}. \tag{116}$$

If

$$S'(\Delta_0) = u_\Delta(T) - \Psi_X(X(T))u_X(T) \neq 0, \tag{117}$$

the initial qualification value is locally unique and varies differentiably with nearby boundary data and smooth primitives. Transversal regime switches use the corresponding piecewise variational equations.

Proof. Initial-value uniqueness follows from the local Lipschitz field, and differentiation gives (116). Applying the chain rule to (115) gives (117). By the implicit-function theorem, a unique initial jump value exists near the reference solution.

Across a transversal endogenous research boundary, the field is continuous and its derivative can be integrated on each side with continuous variation u. For a first-availability event $A = A_M$, the field can change discretely, while its positive frontier drift crosses the event in one direction. We write $f = (H_1, F, G)^\mathsf{T}$ and $n = (0, 1, 0)^\mathsf{T}$. Differentiating the event condition gives the change in its date as $\mathrm{d}t_* = -n^\mathsf{T}u^-/(n^\mathsf{T}f^-)$, where $n^\mathsf{T}f^- = F^- > 0$. Comparing the perturbed path at a common calendar date then gives

$$u^+ = \left[I + \frac{(f^+ - f^-)n^\mathsf{T}}{n^\mathsf{T}f^-} \right] u^-. \tag{118}$$

Inserting this expression between the smooth segments preserves the shooting calculation and the implicit-function argument. □

For implementation of this criterion, the career entries of the Jacobian are

$$\begin{aligned} H_{1E} &= \widehat{M}_E - x - \delta, \quad H_{1A} = \widehat{M}_A, \quad H_{1\Delta} = \widehat{M}_\Delta + (1 - E)/\kappa, \\ G_E &= -W_E, \qquad\qquad G_A = -W_A, \quad G_\Delta = R + x - W_\Delta. \end{aligned} \tag{119}$$

For example, $\widehat{M}_E = M_E + M_w W_E$ holds the primitive wage argument fixed in M_E. In the human region, the remaining row satisfies

$$\frac{F_E}{F} = -\frac{W_E}{w + q\xi}, \quad \frac{F_\Delta}{F} = -\frac{W_\Delta}{w + q\xi}, \quad \frac{F_A}{F} = \frac{\epsilon_H - \phi}{A} - \frac{W_A}{w + q\xi}. \tag{120}$$

In an interior mixed or machine region, $F_E = F_\Delta = 0$ and $F_A/F = (\epsilon_M - \phi)/A$. When the derivative in (117) vanishes, selection depends on the nonlinear residual. At a tangency to an endogenous boundary, the Lipschitz existence theorem still applies, while derivatives require their one-sided versions.

H.4 Career saddle

Proposition H.3 (Selection at a fixed frontier)**.** *At a stationary point with fixed A and continuously differentiable career fields in a neighbourhood, we write $f(E,\Delta) = \widehat{M} + (1-E)\Delta/\kappa - \delta E$ and $g(E,\Delta) = R\Delta - (W-b) + \Delta^2/(2\kappa)$. Derivative conditions*

$$f_E < 0, \qquad f_\Delta \geq 0, \qquad g_E > 0, \qquad g_\Delta > 0 \tag{121}$$

give one stable and one unstable eigenvalue. Its stable manifold is locally a unique graph $\Delta = \chi(E)$ with negative slope at the stationary point, so each nearby inherited expertise level selects one locally convergent career value. In an interior mixed regime with $g_E = 0$, the conditions $f_E < 0 < g_\Delta$ give the same eigenvalue count and a horizontal stable tangent.

Proof. Under (121), the determinant $f_E g_\Delta - f_\Delta g_E$ is strictly negative, so the eigenvalues are real and have opposite signs. For the negative eigenvalue λ_s, its eigenvector (v_E, v_Δ) satisfies $g_E v_E + (g_\Delta - \lambda_s)v_\Delta = 0$. Hence $v_E \neq 0$ and

$$\frac{v_\Delta}{v_E} = -\frac{g_E}{g_\Delta - \lambda_s} < 0.$$

By the stable-manifold theorem, a one-dimensional local manifold has that tangent. Its nonzero projection on E gives the stated graph and selection. When $g_E = 0$, the Jacobian is triangular with eigenvalues $f_E < 0 < g_\Delta$ and stable tangent $(1, 0)$. □

For our strict-margin diagnostic portfolio, $\widehat{M}_E = \widehat{M}_\Delta = W_\Delta = 0$. Thus $f_E = -x - \delta < 0$, $f_\Delta = (1-E)/\kappa > 0$, and $g_\Delta = R + x > 0$. Strictly decreasing wages give the first case of Proposition H.3, and a flat mixed wage gives the second. Under the fixed-contact saturation specialization $M = (1-E)m$, its first derivative becomes $f_E = -m - x - \delta < 0$, preserving the result.

H.5 Announcements and training

Equal packet costs reveal the wage effect of relative research productivity. For the elasticities $\epsilon_H = -Aa'_H/a_H$ and $\epsilon_M = -Aa'_M/a_M$, direct differentiation gives

$$\frac{\mathrm{d}\overline{w}}{\mathrm{d}\log A} = (\overline{w} + q\xi)(\epsilon_H - \epsilon_M). \tag{122}$$

Advancing frontier productivity lowers the mixed wage exactly where the complete machine method improves faster than the complete human method. For the serial research technology, these elasticities approach p_M and p_H, respectively. With $p_M > p_H$, the mixed wage eventually declines, while an order over the full comparison interval gives a wage-path comparison throughout that interval.

Proposition H.4 (Anticipated machine improvement)**.** *For two finite-horizon joint equilibria with common terminal qualification values, common career parameters, and common inherited expertise at date zero, the changes in current qualification value and individual formal study have the sign of*

$$\int_0^T \mathcal{K}(0,s)[w_1(s) - w_0(s)]\,\mathrm{d}s, \tag{123}$$

where $\mathcal{K}$ is the positive kernel in Proposition G.3. Their current aggregate formal-transition difference has the same sign.

For a concrete joint-equilibrium comparison, both economies remain in the interior mixed regime over $[0,T]$ and share initial A and all research requirements before a known adoption date $t_a \in (0,T)$. After that date, an announced improvement gives $a_M^1(A) \leq a_M^0(A)$, with common a_H, B_R, q, ξ. If $\epsilon_M^1(A) \geq \epsilon_H(A)$ over the subsequent frontier interval, the announcement weakly lowers current Δ, individual study, and aggregate formal transitions. Strict future wage reduction on a set of positive measure gives strictness.

Proof. By Proposition G.3, the finite-horizon value difference equals (123) because the terminal difference is zero. Dividing by κ gives the individual training difference. Multiplication by the common positive novice share gives the current aggregate training difference.

In the mixed regime, each frontier follows

$$\dot{A}_i = \frac{B_R A_i^{-\phi}}{q a_M^i(A_i)},$$

independently of E and Δ. Uniqueness gives equal frontier paths before t_a. Afterwards the improved scalar field f_1 is weakly larger than f_0 at every frontier. For $v = (A_0 - A_1)_+$ on any finite compact frontier interval, a Lipschitz constant L for f_0 gives $\dot{v} \le Lv$ almost everywhere. Indeed, when $A_0 > A_1$, their drift difference obeys $f_0(A_0) - f_1(A_1) \le f_0(A_0) - f_0(A_1) \le Lv$. Since $v(t_a) = 0$, integration gives $v = 0$ and hence $A_1 \ge A_0$. At a common frontier the improved method also has weakly lower mixed wage. Equation (122) gives a nonincreasing improved wage schedule, so

$$w_1(s) = \overline{w}_1(A_1(s)) \le \overline{w}_1(A_0(s)) \le \overline{w}_0(A_0(s)) = w_0(s).$$

Before adoption, their common frontier and methods give equal mixed wages. Positive kernel weights now establish the value and training claims. □

Improving the machine-validation coefficient m_R gives one instance of this experiment. Its elasticity condition concerns the complete improved method, and interior mixing requires $w_0 < \overline{w} < w_1$ along both paths. Positive margins in a compact benchmark interval preserve these regime inequalities under sufficiently small changes. Future aggregate training also depends on the novice population, so the common initial headcount has an explicit role in the proposition's aggregate statement.

For varying computation prices with fixed ξ, the wage identity becomes

$$\dot{\overline{w}} = \frac{\dot{q}}{q}\overline{w} + (\overline{w} + q\xi)(\epsilon_H - \epsilon_M)\frac{\dot{A}}{A}. \tag{124}$$

Thus a time-varying compute price enters the wage condition together with relative method productivity.

Proposition H.5 (Announced availability at a fixed frontier)**.** *For the diagnostic portfolio benchmark, hold A fixed and take a common exogenous practice path $M(t)$. Wage schedules $W_i(t, E)$ are independent of Δ, continuously differentiable, nonincreasing in E, and satisfy $W_1(t, E) \le W_0(t, E)$. Two feasible finite-horizon career paths with common initial expertise and terminal qualification value then satisfy $\Delta_1(0) \le \Delta_0(0)$. Current individual and aggregate formal training have the same ordering. Strict wage-schedule reduction along the reference state on a set of positive measure gives strictness.*

Proof. For differences $y = E_1 - E_0$ and $z = \Delta_1 - \Delta_0$, exact subtraction gives

$$\dot{y} = -ay + \beta z, \qquad \dot{z} = cy + dz + e, \qquad y(0) = 0, \quad z(T) = 0, \tag{125}$$

where $a = \delta + \Delta_1/\kappa > 0$, $\beta = (1 - E_0)/\kappa > 0$, $d = R + (\Delta_1 + \Delta_0)/(2\kappa) > 0$, and

$$c = -\int_0^1 W_{1,E}(t, E_0 + s(E_1 - E_0))\, \mathrm{d}s \ge 0, \qquad e = W_0(t, E_0) - W_1(t, E_0) \ge 0.$$

We define p by the terminal Riccati equation

$$\dot{p} = c + (d + a)p - \beta p^2, \qquad p(T) = 0.$$

After reversing time, $r(s) = -p(T-s)$ has drift $c-(d+a)r-\beta r^2$ and initial value zero. Its nonnegative half-line is invariant, and bounded coefficients and the negative quadratic term keep its solution finite. Hence $p \leq 0$. Substitution of $v = z - py$ into (125) gives $\dot{v} = (d-\beta p)v + e$ and $v(T) = 0$. With $y(0) = 0$, variation of constants yields

$$z(0) = -\int_0^T \exp\left[-\int_0^s (d-\beta p)\,\mathrm{d}u\right] e(s)\,\mathrm{d}s \leq 0. \tag{126}$$

Positive weights give strictness, and the common initial novice share gives the training comparisons. □

At a frozen frontier, first availability of autonomous research at a known future date produces the pointwise wage-schedule reduction through (107). Proposition H.5 includes the resulting endogenous career adjustment. Proposition H.4 provides its counterpart with frontier dynamics under an announced improvement of an available method. In the general joint availability comparison, the future wage path reflects changes in expertise, practice credits, frontier productivity, and regime dates, whose combined sign is given by (123).

On a fixed availability region, or across common predetermined calendar dates of coefficient changes, a differentiable local joint comparison indexed by ϑ records the same effect in derivatives. We write $y = \partial_\vartheta X$ and $z = \partial_\vartheta \Delta$, with common initial states and constant terminal qualification value. Along the branch,

$$\dot{y} = H_X y + H_\Delta z + H_\vartheta, \qquad \dot{z} = G_X y + G_\Delta z - W_\vartheta.$$

Where the continuation value has a differentiable state graph, its row derivative P satisfies

$$\dot{P} = G_X + G_\Delta P - PH_X - PH_\Delta P, \qquad P(T) = 0.$$

Indeed, differentiating the state graph along the field and matching the two variational equations gives this identity. Substitution of $r = z - Py$, followed by integration from the common terminal value, yields

$$z(0) = \int_0^T \exp\left[-\int_0^s (G_\Delta - PH_\Delta)\,\mathrm{d}u\right] [W_\vartheta + PH_\vartheta](s)\,\mathrm{d}s. \tag{127}$$

Thus $W_\vartheta + PH_\vartheta \leq 0$ throughout the interval is a sufficient signed condition, with strictness on a positive-measure set. Its PH_ϑ term values the changes in learning and frontier drift induced by the announcement. Existence of this state graph throughout the interval requires the corresponding remaining-horizon shooting derivatives to stay nonzero.

H.6 Completion and continued work

When an autonomous research program has $\mu, \zeta > \phi + 1$, its remaining production time is

$$\mathcal{T}_R(A) = \frac{q}{B_R}\int_A^\infty u^\phi a_M(u)\,\mathrm{d}u. \tag{128}$$

Along an autonomous interval, differentiation gives $\mathrm{d}\mathcal{T}_R/\mathrm{d}t = -1$. We give this finite boundary an explicit economic completion. At zero remaining time the program publishes its complete software library, its research budget retires, and the frontier state is replaced by an absorbing completed-program state. Under this interpretation, the library benefit is bounded and limiting service resources are finite. Residual implementation, site-specific verification, and responsibility for customer systems continue to use portable engineering expertise.

In this continuation occupation, a qualified worker produces $w_\infty > b$ units of the numeraire per hour and a novice produces b. Competitive constant-return production supports those wages. Formal study and obsolescence retain their original technologies, and diagnostic practice is zero. For a constant flow C_∞ from the completed library and independent endowment, the resource flow is

$$b(1-E) + w_\infty E - \frac{\kappa}{2}(1-E)x^2 + C_\infty. \tag{129}$$

Proposition H.6 (Economic continuation)**.** *Within the continuation occupation, qualification value, study intensity, and private values are*

$$\begin{aligned} \Delta_\infty &= \kappa\left[\sqrt{R^2 + 2(w_\infty - b)/\kappa} - R\right], \qquad x_\infty = \Delta_\infty/\kappa, \\ V_{N,\infty} &= \frac{b + \Delta_\infty^2/(2\kappa)}{\rho}, \qquad\qquad V_{E,\infty} = V_{N,\infty} + \Delta_\infty. \end{aligned} \tag{130}$$

For an inherited share at the completion date T_∞, its subsequent path and social continuation value are

$$E(t) = \frac{x_\infty}{\delta + x_\infty} + \left[E(T_\infty) - \frac{x_\infty}{\delta + x_\infty}\right] e^{-(\delta + x_\infty)(t - T_\infty)}, \tag{131}$$

$$\Phi_\infty(E) = V_{N,\infty} + \Delta_\infty E + \frac{C_\infty}{\rho}, \qquad \Phi'_\infty(E) = \Delta_\infty. \tag{132}$$

These values specify the private and social terminal conditions for the research phase.

Proof. Stationarity of the worker equations gives $R\Delta_\infty + \Delta_\infty^2/(2\kappa) = w_\infty - b$, whose positive root and optimal study intensity are (130). Its novice Bellman equation gives $V_{N,\infty}$, and adding the qualification difference gives $V_{E,\infty}$. Substitution of constant x_∞ into $\dot{E} = (1 - E)x_\infty - \delta E$ gives (131).

For the social value, substitute (132) into the stationary resource Bellman equation with flow (129). Its derivative in the study choice is $(1 - E)(\Delta_\infty - \kappa x)$, giving $x = x_\infty$. Its constant term is $b + \Delta_\infty^2/(2\kappa) + C_\infty$, while the coefficient on E is

$$w_\infty - b - \frac{\Delta_\infty^2}{2\kappa} - \delta\Delta_\infty = \rho\Delta_\infty.$$

These terms equal $\rho\Phi_\infty(E)$. Concavity of the formal cost gives the supporting Bellman inequality for every feasible study policy. Integration discounts the bounded terminal value to zero, verifying the social optimum and its marginal qualification value. □

Proposition H.7 (Career continuity at completion)**.** *When an autonomous terminal phase begins at (t_*, A_*), its completion date is*

$$T_\infty = t_* + \frac{q}{B_R}\left[\frac{A_*^{\phi - \mu + 1}}{\mu - \phi - 1} + \frac{m_R A_*^{\phi - \zeta + 1}}{\zeta - \phi - 1}\right]. \tag{133}$$

Along its frontier path, cleared wages and practice are uniformly bounded and Lipschitz in (E, Δ) on a compact feasible career rectangle, with $b \le W \le W_{\max}$. Machine-only procurement holds throughout that rectangle for $A \ge A_*$. If the remaining duration obeys the state-margin and contraction bounds of Proposition H.1, using the single inherited career state E, there is a unique career equilibrium through T_∞ with terminal values (130)*. Joining it to Proposition H.6 defines worker values and expertise at every future date.*

For finite caps $K > A_*$, use the same continuation occupation at the first arrival date T_K of frontier K, with common inherited $(E(t_*), A_*)$. As $K \to \infty$, the career paths converge uniformly on their common precompletion intervals to the completed economy, with error of order $T_\infty - T_K$.

Proof. Integration of the autonomous research law gives (133), and its positive integrand gives $T_K \uparrow T_\infty$. Its frontier path is independent of E and Δ, providing a common deterministic time argument for the career fields. Uniform bounds and Lipschitz constants allow the proof of Proposition H.1 to be applied through the finite endpoint with terminal Δ_∞. Values assigned to the coefficients at that single endpoint leave the integral equations unchanged. Bounded career derivatives give continuous limits of E and Δ, while (114) with terminal $V_{N,\infty}$ supplies the full private values. Afterwards Proposition H.6 defines the remaining career.

For convergence, we denote the completed value by Δ^{∞} and bound the precompletion career drift by $|G| \leq C_G$. Its terminal condition gives

$$|\Delta^{\infty}(T_K) - \Delta_{\infty}| \leq C_G(T_{\infty} - T_K).$$

On $[t_*, T_K]$, the cap economy and the restriction of the completed economy have identical primitive fields and inherited expertise. Their terminal values differ by the preceding bound. For a common trial wage path, the two backward Riccati solutions differ by at most the terminal difference because their discount coefficient is at least R. If $\chi < 1$ is the uniform contraction bound obtained from $T_{\infty} - t_*$, the fixed-point estimate yields

$$\sup_{t_* \leq t \leq T_K} \max\{|E^K(t) - E^{\infty}(t)|, |\Delta^K(t) - \Delta^{\infty}(t)|\} \leq \frac{C_G}{1-\chi}(T_{\infty} - T_K). \tag{134}$$

During $[T_K, T_{\infty}]$, both career drifts are bounded, preserving the same order of endpoint error. Afterwards (131) contracts the expertise difference. Bounded integrands in (114) give convergence of both worker values as well. □

With continued implementation work, portable expertise has an economic value after the research program ends. Its wage is a stated primitive of that continuation. Different choices alter Δ_{∞} through (130). For a common precompletion wage path and two terminal qualification values, direct subtraction of the private equations gives

$$\Delta_1(t) - \Delta_0(t) = \exp\left[-\int_t^T \left(R + \frac{\Delta_1 + \Delta_0}{2\kappa}\right) \mathrm{d}s\right] [\Delta_1(T) - \Delta_0(T)]. \tag{135}$$

With nonnegative values bounded by D and $t \leq T \leq T_{\infty} < \infty$, the weight in (135) is bounded below by

$$e^{-(R+D/\kappa)(T_{\infty}-t)} > 0.$$

Future work after completion continues to affect current study through this positive weight. Our cap sequence preserves the same residual occupation, teaching technology, obsolescence, and portability rule, which gives the limiting career comparison a common economic continuation.

I Research welfare

I.1 Resource account

Our welfare comparison retains privately owned engineering expertise, competitive service firms, and the public fund's mandate to purchase completed research packets. Qualified engineers can supply customer services or generate and validate research in the same technical field, while novices purchase diagnostic placements and formal instruction. Research results become available for common use. Funding from an independently endowed numeraire sector supplies the fund's lump-sum contribution, domestic compute suppliers use resources equal to their receipts at price q, and teaching uses the numeraire resources charged as tuition. We assign equal welfare weights to transfers among these participants and evaluate the resulting resource flows.

For customer mode shares π, aggregate formal transitions I, research expert hours L_R, and autonomous computation Q_M, we write competitive service surplus before the cost of expert time as

$$\mathcal{S}_C(\pi, A) = \int \sum_d \pi_d(z)\{U_d(z, A) - b j_d(z, A)\} \, \mathrm{d}\mu_Z(z). \tag{136}$$

Mode primitives depend on the frontier through the customer interface and the specified technologies. In this benchmark they are independent of E. With diagnostic flow $M(\pi, A)$, expert service demand $D_S(\pi, A)$, and basic demand $J(\pi, A)$, physical feasibility and public finance require

$$\begin{aligned} \dot{E} &= M(\pi, A) + I - \delta E, \\ \dot{A} &= A^{-\phi} \left\{ \frac{L_R}{a_H(A)} + \frac{Q_M}{a_M(A)} \right\} = F, \\ D_S(\pi, A) + L_R = E, \qquad & J(\pi, A) < 1 - E, \\ (w + q\xi)L_R + qQ_M &= B_R, \\ I, L_R, Q_M \geq 0, \qquad \pi_d \geq 0, \qquad & \sum_d \pi_d = 1. \end{aligned} \tag{137}$$

With an unavailable machine route, procurement has $Q_M = 0$. Procurement selects the cheapest complete method, with both methods used at equal packet costs, while customer choices maximize $U_d - b j_d - w e_d + \Delta m_d$ and workers choose $I = (1 - E)\Delta/\kappa$. These decisions specify the equilibrium allocation whose welfare we evaluate.

With independent numeraire endowment Y, the current resource flow is

$$G = Y + b(1 - E) + \mathcal{S}_C(\pi, A) - \frac{\kappa I^2}{2(1 - E)} - q(\xi L_R + Q_M). \tag{138}$$

Research wages and the lump-sum contribution cancel when participant accounts are added. Compute and teaching remain resource costs, while expert feasibility records the professional hours allocated to research. By counting these resources, we value the fund through its physical inputs and the future use of its research, with B_R determining procurement through (137).

On a finite interval $[t, T]$ of finite states, a specified equilibrium continuation gives

$$V(t, E, A) = \int_t^T e^{-\rho(s-t)} G_s \, \mathrm{d}s + e^{-\rho(T-t)} \Phi(E_T, A_T). \tag{139}$$

Here Φ is the resource value of the terminal continuation economy, whose private qualification difference is $\overline{\Delta}(E, A)$. Our derivative results use a regular equilibrium branch with private value $\Delta = \mathfrak{d}(t, E, A)$, wage $w = w(t, E, A)$, positive Δ, and slack novice time. Within each active research regime the branch and V are twice continuously differentiable on an open state region. Its qualification and frontier fields are

$$f = M + (1 - E)\frac{\Delta}{\kappa} - \delta E, \qquad F = \dot{A}, \qquad \lambda = V_E, \qquad \Theta = V_A. \tag{140}$$

Derivatives with respect to E follow this branch at fixed time and frontier, so they include the response of private study and customer selection. Across finitely many regime crossings, we use continuous V_E and $\mathfrak{d}$, piecewise continuous integrable derivatives, and transitions that preserve states and carry zero separate payoff. These conditions permit the integral identities below to pass across the crossings.

We evaluate two local policy controls, a switch of a small class of cases between customer preparation modes and a subsidy per successful formal transition. Each intervention is announced when it begins, operates for a short interval, and is followed by the same equilibrium continuation rule from the resulting states. Lump-sum transfers from the endowed sector finance the subsidy or return any levy receipts. Research procurement continues to follow its stated mandate. In this comparison, V is the resource value of that institution and its continuation decisions.

I.2 Qualification values

Proposition I.1 (Public research and the qualification wedge)**.** *Along the regular branch, the difference $d_V = \lambda - \Delta$ satisfies*

$$\dot{d}_V = (\rho - f_E)d_V - \Gamma_E, \qquad \Gamma_E = L_R w_E + \Theta F_E. \tag{141}$$

With the positive weight

$$K_E(t,s) = \exp\left\{-\int_t^s [\rho - f_E(u)]\,\mathrm{d}u\right\}, \tag{142}$$

its finite-horizon representation is

$$\begin{aligned} d_V(t) = K_E(t,T)\{\Phi_E(E_T,A_T) - \overline{\Delta}(E_T,A_T)\} \\ + \int_t^T K_E(t,s)\Gamma_E(s)\,\mathrm{d}s. \end{aligned} \tag{143}$$

At a strict human corner, we define $h = w + q\xi$ and the gross social value per research hour by

$$\Pi = \Theta \frac{A^{-\phi}}{a_H(A)}. \tag{144}$$

Then

$$\Gamma_E = (\Pi - h)(L_R)_E. \tag{145}$$

In the purchased diagnostic portfolio with service demand independent of Δ, the derivative is

$$(L_R)_E = \frac{B_R}{h^2 \mathcal{J}} > 0, \qquad \mathcal{J} = -D_{S,w} + \frac{B_R}{h^2}. \tag{146}$$

Within an interior mixed allocation or a strict autonomous allocation, we have $\Gamma_E = 0$.

Proof. Evaluation of the resource continuation gives

$$\rho V = G + V_t + V_E f + V_A F.$$

Differentiating in E and following the equilibrium trajectory yields

$$\dot{\lambda} = \rho\lambda - G_E - \lambda f_E - \Theta F_E. \tag{147}$$

Customer optimization supplies a local envelope for selected modes,

$$(\mathcal{S}_C)_E = w(D_S)_E - \Delta M_E. \tag{148}$$

For finite menus, every mode with a positive share within a case has the same maximized value of $U - bj - we + \Delta m$. Multiplying that equality by the derivative of each share and adding gives zero because the shares sum to one, which proves (148). Differentiable interior choices give the same equality through their first-order conditions, and integration over smooth marginal-customer choices preserves it. All mode coefficients are held fixed when differentiating in E at the given frontier.

Writing $x = \Delta/\kappa$ gives $I = (1-E)x$. Differentiating the resource and qualification flows along the branch gives

$$\begin{aligned} G_E = &-b + w(D_S)_E - \Delta M_E + \frac{\kappa x^2}{2} - (1-E)\kappa x x_E \\ &- q\xi(L_R)_E - q(Q_M)_E, \\ f_E = &\, M_E - x + (1-E)x_E - \delta. \end{aligned} \tag{149}$$

Expert clearing and the fixed-budget derivative give

$$(D_S)_E + (L_R)_E = 1, \qquad h(L_R)_E + q(Q_M)_E = -L_R w_E.$$

Using these identities and $\kappa x = \Delta$ in (149) yields

$$G_E + \Delta f_E = w - b - \frac{\Delta^2}{2\kappa} - \delta\Delta + L_R w_E. \tag{150}$$

Meanwhile, the private value along the same trajectory obeys

$$\dot{\Delta} = \rho\Delta - \left\{ w - b - \frac{\Delta^2}{2\kappa} - \delta\Delta \right\}.$$

Subtracting this equation from (147) and using (150) proves (141). Multiplication by the integrating factor and backward integration give (143).

At a human corner, $L_R = B_R/h$ and $F = A^{-\phi} L_R / a_H(A)$ imply

$$(L_R)_E = -\frac{L_R}{h} w_E, \qquad \Theta F_E = \Pi (L_R)_E.$$

Their sum gives (145). When service demand is independent of Δ, differentiating $D_S(w, A) + B_R/h = E$ gives (146). In the mixed region, the equal-cost wage and the machine packet cost depend on the frontier alone, giving $w_E = F_E = 0$. In the autonomous region $L_R = 0$ and $F_E = 0$. These identities prove the remaining statements. □

With its term $L_R w_E$, the value comparison records the procurement consequence of a change in expertise. More experts can reduce the fund's wage payments per hour, changing the physical inputs bought with its budget. Future frontier gains enter through ΘF_E, which values their contribution to all subsequent resource uses. Their difference from the private career return depends on both terms in (141).

Corollary I.2 (A positive training-subsidy margin)**.** *Terminal social and private qualification values agree, and every future human research interval satisfies $(L_R)_E \geq 0$ and $\Pi \geq h$. If both inequalities are strict on a set of positive measure, while the intervening research regimes are mixed or autonomous, then $\lambda_t > \Delta_t$ before that set. For a subsidy σ per successful formal transition, the current-value welfare derivative of a short local intervention at $\sigma = 0$ is*

$$\frac{1 - E}{\kappa}(\lambda - \Delta) > 0. \tag{151}$$

When every future research allocation is mixed or autonomous and terminal qualification values agree, $\lambda = \Delta$ throughout the continuation.

Proof. Every weight in (143) is positive. Its terminal term is zero, while (145) is nonnegative and strictly positive on the stated set. Integrating these terms proves the strict value comparison. At the existing private choice, marginal resource expenditure per formal transition is $\kappa I/(1-E) = \Delta$, and the transition adds λ to continuation welfare. With the subsidy, $x = (\Delta + \sigma)/\kappa$, so the derivative of formal transitions is $(1 - E)/\kappa$ and their net contribution is (151). Its financing is a transfer, and tuition accounts for the additional instruction resources. Our short-intervention argument in Proposition I.3 justifies evaluating the leading derivative at the existing Δ. Finally, mixed and autonomous regimes set every integrand in (143) to zero, proving the last statement. □

We can obtain the positive research-value condition from an explicit public payoff. On a compact finite interval of strict human research, we take service and practice primitives and the private terminal qualification difference to be independent of A. Private careers then depend on time and E, with

$$F = g(A) L_R(t, E), \qquad g(A) = \frac{A^{-\phi}}{a_H(A)} > 0.$$

At the terminal date, publicly available research pays a common dividend with present value βA_T, where $\beta > 0$. For the remaining terminal resource value, we use $\Phi_\infty(E)$ from (132), with private qualification difference Δ_∞, so their qualification derivatives agree by the specified continuation occupation. Because the dividend enters both private status values equally, it preserves their difference and the procurement allocation. Here $f_A = G_A = 0$, so differentiating the resource evaluation equation in A and using $\Theta(T) = \beta$ gives

$$\Theta(t) = \beta \exp\left\{-\int_t^T [\rho - F_A(s)]\,\mathrm{d}s\right\}. \tag{152}$$

Thus the explicit condition

$$\beta > \max_{t\in[0,T]} \frac{h(t)}{g(A(t))\exp\{-\int_t^T [\rho - F_A(s)]\,\mathrm{d}s\}} \tag{153}$$

gives $\Pi(t) > h(t)$ throughout the interval. Continuity and positivity on the compact path ensure a finite maximum. Together with $(L_R)_E > 0$, this public-payoff specification satisfies the strict conditions of Corollary I.2, providing an economy in which research's social value raises the return to qualification above its privately received value.

I.3 Customer preparation

Proposition I.3 (Released hours and the learning margin)**.** *At a strict human corner of the purchased diagnostic portfolio, service demand satisfies $D_{S,w} < 0$, and diagnostic flow outside a negligible designated class is locally fixed. Preparing one case saves $j > 0$ basic hours and $e > 0$ expert hours, uses customer resources $r_C \geq 0$, and removes qualification probability $m > 0$, with service benefit preserved. We define*

$$s_C = bj - r_C, \qquad \chi_R = \frac{B_R}{h^2 \mathcal{J}} \in (0,1). \tag{154}$$

Its private gain and current-value welfare derivative are

$$\begin{aligned} P_C &= s_C + we - \Delta m, \\ \mathcal{W}_C &= s_C + we - \lambda m + (\Pi - h)\chi_R e \\ &= P_C - d_V m + (\Pi - h)\chi_R e. \end{aligned} \tag{155}$$

Research absorbs the fraction χ_R of the expert time released. Its remaining fraction expands professional services, so the social value per released expert hour is

$$w + \chi_R(\Pi - h) = (1 - \chi_R)w + \chi_R(\Pi - q\xi). \tag{156}$$

Proof. For a marginal prepared case mass θ, differentiation of expert clearing at fixed (E, A, Δ) and of human procurement gives

$$D_{S,w}\,\mathrm{d}w - e\,\mathrm{d}\theta + \mathrm{d}L_R = 0, \qquad \mathrm{d}L_R = -\frac{B_R}{h^2}\,\mathrm{d}w.$$

It follows that

$$\frac{\mathrm{d}w}{\mathrm{d}\theta} = -\frac{e}{\mathcal{J}}, \qquad \frac{\mathrm{d}L_R}{\mathrm{d}\theta} = \chi_R e, \qquad D_{S,w}\frac{\mathrm{d}w}{\mathrm{d}\theta} = (1 - \chi_R)e. \tag{157}$$

Strictly negative service-demand slope gives $0 < \chi_R < 1$. Other professional services consequently expand their use of expert hours by $(1 - \chi_R)e$. Their marginal resource benefit is $w(1 - \chi_R)e$ by the competitive customer envelope and fixed diagnostic flow. Preparation directly saves s_C, while the extra research uses $q\xi\chi_R e$

computation resources. Qualification drift changes by $-m$ and frontier drift by $A^{-\phi}\chi_R e/a_H(A)$. Summing current resources and their continuation values gives

$$s_C + w(1-\chi_R)e - q\xi\chi_R e - \lambda m + \Pi\chi_R e,$$

which is (155). Private fees value saved expert hours at w, basic hours at b, and learning rights at Δ, yielding P_C. Rearrangement proves (156).

For the timing argument, we apply the previously unannounced case change with intensity θ on $[t, t+\varepsilon]$, then resume the same equilibrium continuation from the resulting states. We denote its current resource value by $\mathcal{V}_t^{\theta,\varepsilon}$. Its terminal private difference for this short interval is $\mathfrak{d}(t+\varepsilon, E_{t+\varepsilon}, A_{t+\varepsilon})$. Applying the short-horizon contraction in Proposition H.1 to the perturbed interval gives continuous dependence with uniform local bounds and state changes of order $\theta\varepsilon$, so differentiability of the continuation gives a terminal private-value change of the same order. During the interval, the wage forcing is of order θ. Integrating the private value equation backward then gives a qualification-value change of order $\theta\varepsilon$ throughout that interval.

Its effect on training and customer demand integrates to order $\theta\varepsilon^2$. Our leading instantaneous market calculation consequently uses the original Δ, with later effects of the changed states valued through V_E and V_A. Differentiating first in θ at zero and then sending the interval length to zero yields

$$\lim_{\varepsilon\downarrow 0}\frac{1}{\varepsilon}\left.\frac{\partial \mathcal{V}_t^{\theta,\varepsilon}}{\partial\theta}\right|_{\theta=0} = \mathcal{W}_C. \tag{158}$$

Applying this reasoning to the short training subsidy in Corollary I.2 gives its leading derivative, since the direct resource and transition changes have bounded derivatives and the continuation follows the same rule. □

Corollary I.4 (Research expansion with a welfare loss)**.** *At the comparison in Proposition I.3, the inequalities* $\lambda > \Delta$ *and*

$$K_C = d_V m - (\Pi - h)\chi_R e > 0 \tag{159}$$

define a region where privately chosen preparation has a strictly negative welfare derivative exactly when

$$\Delta m - we < s_C < \lambda m - [w + (\Pi - h)\chi_R]e. \tag{160}$$

Throughout this interval of positive length K_C, human research expands by $\chi_R e$ per prepared case.

Proof. Private choice requires $P_C > 0$ and a welfare loss requires $P_C < K_C$ by (155). These inequalities give (160), whose endpoint difference is K_C. Research expansion follows from (157).

For a concrete symbolic construction with nonnegative customer cost, we choose $j > 0$ and $m \in (0, 1]$ satisfying

$$bj > \frac{\lambda + \Delta}{2}m.$$

Such a choice is available by taking a sufficiently small positive qualification probability. We choose $e > 0$ sufficiently small that $K_C > 0$ and

$$s_C^* = \Delta m - we + \frac{K_C}{2} < bj, \qquad r_C = bj - s_C^* > 0.$$

These inequalities follow by continuity as e approaches zero, since K_C approaches $d_V m > 0$ and s_C^* approaches $(\lambda + \Delta)m/2$. Substitution gives $P_C = K_C/2 > 0$ and $\mathcal{W}_C = -K_C/2 < 0$, together with the strictly positive research increase $\chi_R e$. Scaling the designated case class preserves the benchmark's strict resource and procurement margins. □

In this local experiment, a levy of K_C per prepared case equates the customer's gain with the resource welfare derivative, with receipts returned through the numeraire sector. Corollary I.2 supplies another positive local policy direction when the social qualification value exceeds its private price. Each result values the change at the same continuation institution.

At an interior mixed allocation, a released expert hour raises human research by one hour and reduces autonomous computation by h/q. Equal packet costs keep total research output fixed. Net compute-resource saving is $h - q\xi = w$, yielding $\mathcal{W}_C = s_C + we - \lambda m$ for the same diagnostic portfolio. At a strict autonomous corner, released expert hours expand other professional services and research output stays fixed, giving the same expression. Their qualification value difference still follows (143), which includes subsequent human research intervals.

For a general differentiable customer portfolio, the wage response can also change other diagnostic opportunities. Holding Δ fixed in the leading instantaneous comparison, their transition response is $M_w(-e/\mathcal{J})$. Applying the competitive service envelope gives the extended human-regime expression

$$\mathcal{W}_C = P_C - d_V m - d_V M_w \frac{e}{\mathcal{J}} + (\Pi - h)\chi_R e. \tag{161}$$

Thus (160) uses the purchased diagnostic portfolio to hold the additional practice response at zero, while (161) retains its contribution in the general local comparison.

I.4 Eligible learners

When a mode's diagnostic yield depends directly on the number of eligible novices, the social qualification derivative also values this acquisition margin. With $m_d = m_d(E, A)$ and the remaining mode primitives independent of E, we define the derivative at fixed customer shares by

$$M_E^{\text{direct}} = \int \sum_d \pi_d(z)\partial_E m_d(z, E, A)\, \mathrm{d}\mu_Z(z). \tag{162}$$

Differentiation of selected mode shares in the competitive envelope now gives

$$(\mathcal{S}_C)_E = w(D_S)_E - \Delta\{M_E - M_E^{\text{direct}}\}.$$

Applying (150) consequently adds $\Delta M_E^{\text{direct}}$ to its right-hand side. Repeating the subtraction of the private value equation yields (141) and (143) with

$$\Gamma_E = L_R w_E + \Theta F_E + \Delta M_E^{\text{direct}}. \tag{163}$$

For random contact with an eligible novice, $m_d(E, A) = (1-E)\ell_d(A)$ gives the additional term $-\Delta m(\pi)$, where $m(\pi) = \int \sum_d \pi_d \ell_d \, \mathrm{d}\mu_Z$ as in (58). When the expert population expands, fewer novices remain eligible for those contacts, so this acquisition effect enters the social-private comparison even during complete research autonomy. Equation (163) gives the corresponding value comparison for that extension.

J Growth

J.1 Expertise persistence

Our growth analysis begins with the people whose qualifications supply research labor. Along any feasible career path with $M \geq 0$ and $\Delta \geq 0$, the workforce equation gives

$$E(t) \geq E(0)e^{-\delta t}. \tag{164}$$

Thus every bounded time interval retains a positive qualified population, even when customer preparation removes all diagnostic practice. Over an infinite horizon, continued human research expenditure provides a stronger bound through the private return to formal training.

Proposition J.1 (Private training and expertise persistence)**.** *For fixed $B_R, b, \kappa, \delta, \rho > 0$, a global differentiable career path satisfies $0 < E < 1$, $\Delta \geq 0$, $M \geq 0$, and*

$$\begin{aligned} \dot{E} &= M + (1-E)\Delta/\kappa - \delta E, \\ \dot{\Delta} &= R\Delta - (w-b) + \Delta^2/(2\kappa), \qquad R = \rho + \delta. \end{aligned} \tag{165}$$

Research purchases $L_R = B_R/(w + s_R)$ qualified hours, where $0 \leq s_R(t) \leq \bar{s}_R < \infty$, and clearing gives $E = D_S + L_R$ with $D_S \geq 0$. For sufficiently small $e \in (0,1)$, the constants

$$d = \frac{\kappa\delta e}{1-e}, \qquad a = \frac{B_R}{e} - \bar{s}_R - b - Rd - \frac{d^2}{2\kappa} \tag{166}$$

satisfy $a > 0$, and the qualified population obeys

$$E(t) \geq E_{\min} := \min\{E(0), e\} \exp(-\delta d/a) > 0 \qquad (t \geq 0). \tag{167}$$

Proof. Nonnegative service demand and research clearing give

$$w + s_R = \frac{B_R}{L_R} \geq \frac{B_R}{E}.$$

Within the region $E \leq e$, $0 \leq \Delta \leq d$, the private value equation consequently implies

$$\dot{\Delta} \leq Rd + b + \bar{s}_R - B_R/e + d^2/(2\kappa) = -a.$$

Because $d \to 0$ while $B_R/e \to \infty$ as $e \downarrow 0$, a choice with $a > 0$ exists.

At a downward crossing of $E = e$, the workforce equation and $M \geq 0$ imply $\Delta \leq d$. While the path remains below e, its value then decreases at rate at least a, which prevents exit through $\Delta = d$. Global nonnegativity of Δ forces a return to $E = e$ within at most d/a time units. During that excursion, $\dot{E} \geq -\delta E$, giving $E \geq e\exp(-\delta d/a)$.

For an initial point with $E(0) < e$ and $\Delta(0) > d$, the workforce satisfies

$$\dot{E} \geq (1-e)d/\kappa - \delta e = 0$$

while $E \leq e$ and $\Delta \geq d$. Its population is nondecreasing until it reaches e or enters the preceding region. Starting inside that region gives an initial excursion with population at least $E(0)$, whose duration is at most d/a. Applying the same bound to every subsequent excursion proves (167). □

In the original human method, $s_R = q\xi$ is the computation expense per researcher hour. Proposition J.1 permits $M \equiv 0$ and places no prior upper bound on Δ. Scarcity of qualified people raises the wage supported by the research budget, and the resulting career return sustains formal entry.

To turn this population bound into a wage bound, we use a common service-demand envelope

$$D_S(w, c, \Delta) \leq \overline{D}(w), \qquad \lim_{w\to\infty} \overline{D}(w) = 0. \tag{168}$$

Choosing W with $\overline{D}(W) + B_R/W < E_{\min}$ gives $w \leq W$ whenever human research clears, since both terms in total labor demand would otherwise sum to less than $E_{\min}$. With both research methods available, the upper human-corner wage w_1 gives the same bound whenever a positive expertise floor is available. Our envelope restricts the demand generated by continuing customer improvements. For example, $D_S = C(c)w^{-\varepsilon}$ with $\varepsilon > 0$ satisfies it whenever $C(c)$ remains bounded as customer capability increases. Continuing growth in discovered services can require a different demand bound.

J.2 Production bounds

For frontier productivity $A \geq 1$, the complete-method exponents are

$$p_H = \min\{\alpha, \nu\}, \qquad p_M = \min\{\mu, \zeta\}. \tag{169}$$

Positive constants a_-, a_+, b_-, b_+ satisfy

$$a_- A^{-p_H} \leq a_H(A) \leq a_+ A^{-p_H}, \qquad b_- A^{-p_M} \leq a_M(A) \leq b_+ A^{-p_M}. \tag{170}$$

Factoring A^{-p_H} from $A^{-\alpha} + v_R A^{-\nu}$ leaves a term with exponent zero and a positive coefficient, while its sum is at most $1 + v_R$, proving the human bounds. Applying the same factorization with coefficients $1, m_R$ proves the machine bounds. Starting below a frontier value of one changes only the constants over a compact initial interval.

We evaluate the production law on its maximal interval $[0, T_{\max})$ of finite states along a selected feasible market and career path. Continuation at a bounded frontier means that a finite endpoint with bounded A admits continuation of the other states and market clearing in the stated feasible region. Requiring this property distinguishes frontier divergence from failure of an occupational or market feasibility condition.

Proposition J.2 (Frontier growth under bounded input prices)**.** *Along a path with the preceding continuation property, the effective human research price satisfies*

$$0 < h_- \leq w + q\xi \leq h_+ < \infty.$$

Human research remains available, and the autonomous route either remains unavailable or stays available beyond a finite frontier threshold. With $p = p_H$ in the first case and $p = \max\{p_H, p_M\}$ in the second, positive constants C_-, C_+ give

$$C_- A^{p-\phi} \leq \dot{A} \leq C_+ A^{p-\phi} \tag{171}$$

after a finite initial segment. For this production law, $T_{\max} < \infty$ exactly when $p > \phi + 1$, and a finite endpoint has $A(t) \to \infty$. At $p = \phi + 1$, frontier productivity has exponential order. At $p < \phi + 1$, it has power order $t^{1/(1+\phi-p)}$, with proportional growth tending to zero. Maintaining only $w + q\xi \geq h_- > 0$ still rules out finite-time frontier divergence when $p \leq \phi + 1$.

Proof. Combining (170) with the two price bounds places human packet cost between positive constants times A^{-p_H}. Machine packet cost has the corresponding bounds with exponent p_M. When both methods are available, their minimum cost lies between positive constants times $A^{-\max\{p_H, p_M\}}$. Substitution into $F = B_R A^{-\phi}/k_*$ proves (171). On the compact frontier interval before a finite entry threshold, the same bounds give a positive minimum drift, so continued paths reach that threshold in finite time.

Writing $\beta = p - \phi$, integration gives, for $\beta < 1$,

$$A(t_0)^{1-\beta} + (1-\beta)C_-(t - t_0) \leq A(t)^{1-\beta} \leq A(t_0)^{1-\beta} + (1-\beta)C_+(t - t_0).$$

These inequalities give the power order, global finite values under continuation, and

$$0 < \dot{A}/A \leq C_+ A^{\beta-1} \longrightarrow 0.$$

At $\beta = 1$, integrating $C_- \leq \dot{A}/A \leq C_+$ gives positive exponential lower and upper bounds. At $\beta > 1$,

$$A(t)^{1-\beta} \leq A(t_0)^{1-\beta} - (\beta - 1)C_-(t - t_0),$$

whose right side reaches zero in finite time. Continuation at bounded frontier states implies a finite maximal interval and divergence of A at its endpoint. Finally, the lower price bound preserves a positive lower packet-cost bound and hence the upper drift bound. Scalar comparison remains finite at every finite date for $\beta \leq 1$. □

Corollary J.3 (Growth with vanishing practice)**.** *In a human-only research economy with fixed B_R, q, ξ, $M \geq 0$, $w \geq b > 0$, the service envelope* (168)*, and continuation at bounded frontier states, the classification in Proposition J.2 holds with $p = p_H$. Practice may converge to zero or equal zero throughout.*

Proof. For a global feasible path, Proposition J.1 and the demand envelope give a uniform upper wage bound, while $w \geq b$ gives the lower price bound. Proposition J.2 applies to that path. At a hypothetical finite endpoint $T_{\max}$, inequality (164) gives the uniform preterminal floor $E(0)e^{-\delta T_{\max}}$, and the envelope again bounds wages on the entire preterminal interval. For $p_H \leq \phi + 1$, the upper drift bound and continuation rule exclude that finite endpoint. For $p_H > \phi + 1$, a hypothetical global path would have the global bounds just established, and the lower drift bound would force finite-time divergence. These two arguments prove the classification. □

J.3 Customer losses

For a customer interface whose occupational effects settle after a finite capability, the aggregate diagnostic transition flow becomes $\overline{M} \geq 0$, and the service schedule becomes a fixed $D(w)$ independent of Δ. Customer capability itself can continue to increase. Complete removal of practice is the case $\overline{M} = 0$, while strict choice margins in the diagnostic portfolio support this specialization over its relevant price range.

We retain human research, fixed B_R, q, ξ, and a continuously differentiable service schedule that is strictly decreasing wherever used. Its clearing equation

$$E = D(w) + \frac{B_R}{w + q\xi} \tag{172}$$

defines a strictly decreasing continuously differentiable inverse wage $w(E)$ on the feasible population region. In the positive-premium region, stationary private training and occupational supply at a given wage are

$$x(w) = \sqrt{R^2 + \frac{2(w-b)}{\kappa}} - R, \qquad S_{\overline{M}}(w) = \frac{\overline{M} + x(w)}{\delta + x(w)}, \qquad 0 \leq \overline{M} < \delta. \tag{173}$$

For $0 \leq \overline{M} < \delta$, supply is strictly increasing in $w > b$, because

$$x_w = \frac{1}{\kappa(R+x)} > 0, \qquad (S_{\overline{M}})_w = \frac{(\delta - \overline{M})x_w}{(\delta + x)^2} > 0.$$

If $S_{\overline{M}}(w) - D(w) - B_R/(w + q\xi)$ changes sign over a positive-premium wage interval, its unique root gives the stationary career allocation in that interval.

Proposition J.4 (Convergence of a settled career economy)**.** *For the settled customer economy above, $0 \leq \overline{M} < \delta$, the service envelope holds, and the inverse wage is continuously differentiable and strictly decreasing on $0 < E < 1$ wherever the path lies. If its positive-premium stationary point (E_*, Δ_*) exists, every global feasible career path with $w > b$ and $\Delta \geq 0$ converges to that point.*

Proof. Proposition J.1 bounds E away from zero, and the envelope supplies a finite wage upper bound $W > b$. We define $D_W > 0$ by

$$RD_W + \frac{D_W^2}{2\kappa} = W - b.$$

Every global finite value path has $\Delta \leq D_W$. Indeed, a value above this root satisfies

$$\dot{\Delta} \geq R\Delta + \Delta^2/(2\kappa) - (W - b) > 0,$$

and comparison with this scalar Riccati equation gives finite-time divergence. Its increasing positive solution reaches any larger fixed level in finite time, after which its quadratic drift gives a finite remaining time to infinity. Such divergence contradicts a global finite value path.

With this bound, workforce accumulation obeys

$$\dot{E} \le \overline{M} + (1-E)D_W/\kappa - \delta E.$$

Because $\overline{M} < \delta$, the comparison equilibrium lies below one. Together with the positive lower population bound, this places the trajectory in a compact region with $0 < E < 1$ and bounded Δ.

At a date with $\dot{E} = 0$, the workforce equation gives

$$\Delta = h_{\overline{M}}(E) := \kappa \frac{\delta E - \overline{M}}{1-E} \ge 0.$$

Differentiating $\dot{E} = \overline{M} + (1-E)\Delta/\kappa - \delta E$ at that date shows that $\ddot{E}$ has the sign of

$$H_{\overline{M}}(E) = R h_{\overline{M}}(E) + h_{\overline{M}}(E)^2/(2\kappa) - [w(E) - b].$$

On $E \ge \overline{M}/\delta$, $h_{\overline{M}}$ strictly increases and w strictly decreases, so $H_{\overline{M}}$ strictly increases and vanishes exactly at E_*. Every local maximum of a nonconstant $E(t)$ consequently lies below E_*, while every local minimum lies above it. Consecutive extrema would require a maximum smaller than its neighboring minimum, contradicting their order along the trajectory. Having a zero derivative at $E = E_*$ also gives $\Delta = \Delta_*$, where uniqueness of the autonomous initial-value problem yields the constant trajectory. Hence each nonconstant path is eventually monotone in E.

Boundedness gives a limit E_∞. Within the compact region, the vector field and its first derivatives are bounded, so $\dot{E}$ is uniformly continuous. Eventual monotonicity and convergence then imply $\dot{E} \to 0$, since a derivative bounded away from zero on a sequence of intervals of uniformly positive length would prevent convergence. Solving the workforce equation for Δ gives $\Delta \to h_{\overline{M}}(E_\infty)$. Its derivative converges to the value-equation field at this limit. Any nonzero limiting derivative contradicts convergence of Δ, so the limit satisfies both stationary equations and equals (E_*, Δ_*). □

Near the stationary point, Proposition H.3 supplies a locally selected convergent career path. Because (172) is independent of frontier productivity, the same career path supports a changing frontier after the occupational effects have settled. For the growth conclusions below, convergence determines the labor quantity that remains after customer preparation and private training have completed their adjustment.

Proposition J.5 (Practice loss and released professional hours)**.** *Two settled customer economies have common $B_R, q, \xi, \rho, \delta, \kappa, b$, aggregate practice flows $0 \le \overline{M}_1 < \overline{M}_0 < \delta$, and service schedules D_0, D_1 satisfying the preceding existence conditions. With w_0 the first economy's stationary wage, $x_0 = x(w_0)$, and $L_i = B_R/(w_i + q\xi)$, stationary research labor satisfies*

$$L_1 \ge L_0 \quad \Longleftrightarrow \quad D_0(w_0) - D_1(w_0) \ge \frac{\overline{M}_0 - \overline{M}_1}{\delta + x_0}. \tag{174}$$

Strict inequalities and equalities correspond on both sides.

Proof. For $i \in \{0, 1\}$, stationary excess supply

$$G_i(w) = S_{\overline{M}_i}(w) - D_i(w) - \frac{B_R}{w + q\xi}$$

is strictly increasing, and $G_0(w_0) = 0$. Evaluating the second economy at the old wage gives

$$G_1(w_0) = \frac{\overline{M}_1 - \overline{M}_0}{\delta + x_0} + D_0(w_0) - D_1(w_0).$$

Its unique root lies at or below w_0 exactly when this expression is nonnegative. Research labor strictly decreases in the wage, which proves the equivalence and its strict and equality versions. □

On the right side of (174), the ratio measures the stationary headcount loss at the original individual training incentive. Service-hour savings above this amount increase the labor purchased by research. Holding the service schedule fixed gives a lower research labor quantity after every practice loss, including a change from positive practice to $\overline{M}_1 = 0$.

Writing

$$a_\infty = \lim_{A\to\infty} A^{p_H} a_H(A) > 0, \qquad K_i = \frac{L_i}{a_\infty}, \tag{175}$$

we obtain a direct frontier consequence.

Corollary J.6 (Long-run customer effects)**.** *Along the global career paths of Proposition J.4, human frontier growth satisfies*

$$\begin{aligned} p_H < \phi + 1 \quad &\Longrightarrow \quad A_i(t)^{1+\phi-p_H} \sim (1+\phi-p_H)K_i t, \\ p_H = \phi + 1 \quad &\Longrightarrow \quad \frac{\dot{A}_i(t)}{A_i(t)} \longrightarrow K_i, \qquad \frac{\log A_i(t)}{t} \longrightarrow K_i. \end{aligned} \tag{176}$$

Thus (174) *determines the direction of the asymptotic level coefficient below the critical exponent and the exponential growth rate at it.*

Proof. Career convergence gives $L_R(t) \to L_i$, while $A^{p_H} a_H(A) \to a_\infty$. Human frontier output consequently satisfies

$$\dot{A}_i \, A_i^{\phi-p_H} \longrightarrow K_i.$$

For $p_H < \phi + 1$, differentiating $A_i^{1+\phi-p_H}$ gives a limit $(1+\phi-p_H)K_i$, and integration proves the first result. At equality, the same expression gives $\dot{A}_i/A_i \to K_i$, whose time average proves the logarithmic limit. □

For stationary career initialization, $L_R = L_i$ throughout and the full frontier arrival relation is

$$t - t_0 = \frac{1}{L_i} \int_{A_0}^{A(t)} u^\phi a_H(u) \, \mathrm{d}u. \tag{177}$$

Every fixed frontier target then arrives earlier exactly when L_i is larger. In the supercritical case, the finite production endpoint also scales as $1/L_i$, with stationary career initialization supporting that comparison. Along a general career transition, the frontier endpoint can arrive before the population approaches its stationary level.

J.4 Human grounding

Complete autonomous validation in the benchmark gives zero local expertise feedback in both the machine and interior mixed regimes at matched frontier productivity. We now require a completed machine packet to use $a_G(A) > 0$ qualified hours for grounding its result in the profession's assessed knowledge, alongside the computation $a_M(A)$ used in machine generation and validation. When this is the sole research method, its unit cost, packet quantity, and labor clearing are

$$k_G = q a_M + w a_G, \qquad Z = \frac{B_R}{k_G}, \qquad E = D_S(w, c, \Delta) + a_G Z. \tag{178}$$

Proposition J.7 (Grounding and expertise feedback)**.** *At fixed A, c, Δ, differentiable demand has $D_{S,w} < 0$. With*

$$\mathcal{J}_G = -D_{S,w} + \frac{B_R a_G^2}{k_G^2} > 0, \tag{179}$$

the grounded frontier $F = A^{-\phi}Z$ satisfies

$$w_E = -\frac{1}{\mathcal{J}_G}, \qquad \frac{F_E}{F} = \frac{a_G}{k_G \mathcal{J}_G} > 0, \qquad \mathrm{d}\log F = \frac{a_G}{k_G \mathcal{J}_G}\left(\mathrm{d}E - \mathrm{d}D_S|_w\right) \tag{180}$$

for matched-frontier customer perturbations holding the research technology and budget fixed.

Proof. Differentiating (178) gives

$$\mathrm{d}E = -\mathcal{J}_G\,\mathrm{d}w + \mathrm{d}D_S|_w.$$

Since $\mathrm{d}\log F = -a_G\,\mathrm{d}w/k_G$ at fixed frontier and research resources, substitution proves all three identities. □

For a grounding requirement $a_G(A) = g_G A^{-\chi}$, with $g_G > 0$ and $\chi \geq 0$, the effective computation overhead per grounding hour is

$$s_R(A) = \frac{q a_M(A)}{a_G(A)}. \tag{181}$$

Whenever $\chi \leq p_M$, this overhead is bounded over $A \geq A_0 > 0$. Research labor equals $B_R/[w + s_R(A)]$, so Proposition J.1 provides an endogenous expertise floor with $M \geq 0$. Under the service envelope and $w \geq b > 0$, wages are bounded and

$$k_G(A) \asymp A^{-\chi}, \qquad \dot{A} \asymp A^{\chi-\phi}. \tag{182}$$

Applying the comparison proof of Proposition J.2 gives finite-time production divergence exactly when $\chi > \phi + 1$, under continuation at bounded frontier states. Hence a grounding stage with $\chi \leq \phi + 1 < p_M$ changes an explosive complete-machine law into a global law. Under independently maintained two-sided wage bounds, the general grounded exponent is $\min\{p_M, \chi\}$, directly from the two positive terms in k_G. When $\chi > p_M$, computation dominates cost and research labor demand can decay with frontier productivity, so the persistence argument requires a separate source of continuing demand for expertise.

If both the original human route and the grounded machine route operate, their equal unit costs instead give

$$w(a_H - a_G) = q(a_M - \xi a_H). \tag{183}$$

For $a_H \neq a_G$, equality fixes the wage at a matched frontier and thus fixes the common packet price. Total packets remain B_R/k_* inside the mixed region while headcount changes alter the route shares, giving $F_E = 0$ there. Grounding restores a positive local feedback at the grounded corner, while equal-cost substitution can absorb population changes inside a mixed allocation.

J.5 Autonomous resources

Proposition J.8 (Autonomous arrival time)**.** *From frontier $A_* > 0$ at date t_*, fixed $B_R, q > 0$ and autonomous procurement give a unique maximal positive solution satisfying*

$$t - t_* = \frac{q}{B_R}\int_{A_*}^{A(t)} \left[u^{\phi-\mu} + m_R u^{\phi-\zeta}\right] \mathrm{d}u. \tag{184}$$

Its frontier diverges in finite time exactly when both $\mu > \phi + 1$ and $\zeta > \phi + 1$, in which case

$$T_\infty - t_* = \frac{q}{B_R}\left[\frac{A_*^{\phi-\mu+1}}{\mu - \phi - 1} + \frac{m_R A_*^{\phi-\zeta+1}}{\zeta - \phi - 1}\right]. \tag{185}$$

Availability of this autonomous route also gives this joint inequality as a sufficient condition for finite-time frontier divergence along any feasible procurement path with continuation at bounded frontier states, including a path using cheaper human packets.

Proof. Positivity and local Lipschitzness of $B_R A^{-\phi}/[q a_M(A)]$ for $A > 0$ give a unique maximal positive solution. Separating variables proves (184). Each nonnegative summand in its improper integral converges exactly when its exponent is below -1, and evaluating those integrals gives (185). When the route is available, $k_* \le q a_M(A)$, so actual frontier drift is at least autonomous drift. Scalar comparison and the continuation property bound its finite endpoint above by the autonomous arrival date. □

We can allow the budget and computation price to change without altering the serial research requirements. For positive continuous functions $B_R(t), q(t)$, we define the cumulative physical research resources

$$S(t) = \int_0^t \frac{B_R(s)}{q(s)}\,\mathrm{d}s, \qquad H(A) = \int_{A_0}^{A} u^{\phi} a_M(u)\,\mathrm{d}u, \qquad H_\infty = \lim_{A\to\infty} H(A). \tag{186}$$

Proposition J.9 (Research resources and frontier growth)**.** *With autonomous procurement and the positive continuous resource paths above, the unique maximal frontier solution satisfies*

$$H(A(t)) = S(t). \tag{187}$$

Its finite endpoint is the first finite date at which $S(t) = H_\infty < \infty$. If $S_\infty := \lim_{t\to\infty} S(t) < H_\infty$, frontier productivity converges to $H^{-1}(S_\infty) < \infty$. Equality $S_\infty = H_\infty < \infty$ reached only at infinite time gives frontier divergence only at infinite time. If $S_\infty = H_\infty = \infty$, the frontier is global and unbounded.

Proof. Positivity of $H'(A) = A^{\phi} a_M(A)$ gives a strictly increasing inverse on $[0, H_\infty)$. Chain differentiation of the frontier equation yields

$$\frac{\mathrm{d}}{\mathrm{d}t} H(A(t)) = \frac{B_R(t)}{q(t)}.$$

Integration gives (187), and $A(t) = H^{-1}(S(t))$ constructs its unique solution up to the stated endpoint. Taking limits through this increasing inverse proves each remaining case. □

For the serial machine technology,

$$b_\infty = \lim_{A\to\infty} A^{p_M} a_M(A) = \begin{cases} 1, & \mu < \zeta, \\ m_R, & \zeta < \mu, > 0. \\ 1 + m_R, & \mu = \zeta \end{cases} \tag{188}$$

Integrating the leading positive power in H' gives the following exact asymptotic relations. For $\gamma = 1 + \phi - p_M > 0$ and an unbounded resource clock,

$$A(t)^{\gamma} \sim \frac{\gamma}{b_\infty} S(t). \tag{189}$$

At $p_M = \phi + 1$, the corresponding relation is

$$\log A(t) \sim S(t)/b_\infty. \tag{190}$$

For $d_M = p_M - \phi - 1 > 0$, approaching the finite resource threshold gives

$$A(t)^{-d_M} \sim \frac{d_M}{b_\infty}\big[H_\infty - S(t)\big]. \tag{191}$$

To see these constants directly, $H'(A) \sim b_\infty A^{\gamma-1}$ below the threshold, $H'(A) \sim b_\infty/A$ at it, and $H_\infty - H(A) \sim b_\infty A^{-d_M}/d_M$ above it.

With constant B_R/q, equation (189) becomes

$$A(t)^{1+\phi-p_M} \sim (1+\phi-p_M)\frac{B_R}{qb_\infty}\,t.$$

At equality, direct substitution into the frontier equation gives

$$\frac{\dot{A}}{A} \longrightarrow \frac{B_R}{qb_\infty}, \qquad \frac{\log A(t)}{t} \longrightarrow \frac{B_R}{qb_\infty}.$$

If $B_R(t)/q(t) \sim r_0 e^{gt}$ with $g > 0$, then $S(t) \sim (r_0/g)e^{gt}$. Equations (189) and (190) give, respectively,

$$\frac{\log A(t)}{t} \longrightarrow \frac{g}{1+\phi-p_M}, \qquad \frac{\log\log A(t)}{t} \longrightarrow g. \tag{192}$$

Integrable declining resource paths instead supply a finite S_∞, whose position relative to H_∞ determines a bounded frontier, a finite endpoint, or divergence at infinite time.

For frontier-indexed resources satisfying $B_R(A) \sim B_0 A^{\beta_B}$ and $q(A) \sim q_0 A^{\beta_q}$, autonomous output satisfies

$$\dot{A} \sim \frac{B_0}{q_0 b_\infty} A^{p_M-\phi+\beta_B-\beta_q}. \tag{193}$$

Separation of variables gives finite-time divergence exactly when $p_M + \beta_B - \beta_q > \phi + 1$, while equality gives exponential order, for the autonomous allocation. With human research, $D_S = 0$, and $\xi = 0$, clearing gives $w = B_R/E$ and

$$\dot{A} = A^{-\phi}E/a_H(A),$$

which is independent of the budget at fixed E, A. More generally, $L_R \le E \le 1$ implies $\dot{A} \le A^{-\phi}/a_H(A)$ for the human route. Budget growth can raise career returns and future entry, while the current physical supply of qualified hours still bounds human packet output.

J.6 Acceleration

For the exact power law $\dot{A} = kA^\beta$, $k > 0$, direct integration gives

$$A(t) = \begin{cases} [A_0^{1-\beta} + (1-\beta)kt]^{1/(1-\beta)}, & \beta \neq 1, \\ A_0 e^{kt}, & \beta = 1, \end{cases} \tag{194}$$

over the interval where the bracket is positive. Its level acceleration and proportional-growth acceleration are

$$\ddot{A} = \beta k^2 A^{2\beta-1}, \qquad g_A = \dot{A}/A = kA^{\beta-1}, \qquad \dot{g}_A = (\beta-1)g_A^2. \tag{195}$$

Thus $0 < \beta < 1$ combines increasing absolute increments with a declining proportional growth rate, while $\beta = 1$ gives constant proportional growth and $\beta > 1$ gives increasing proportional growth with finite-time divergence.

For the serial research methods, local cost elasticities are

$$\epsilon_H(A) = \frac{\alpha A^{-\alpha} + v v_R A^{-\nu}}{A^{-\alpha} + v_R A^{-\nu}}, \qquad \epsilon_M(A) = \frac{\mu A^{-\mu} + \zeta m_R A^{-\zeta}}{A^{-\mu} + m_R A^{-\zeta}}. \tag{196}$$

Along a differentiable human corner with fixed B_R, q, log differentiation gives

$$\frac{\ddot{A}}{\dot{A}} = (\epsilon_H - \phi)g_A - \frac{\dot{w}}{w+q\xi}, \qquad \frac{\dot{g}_A}{g_A} = (\epsilon_H - \phi - 1)g_A - \frac{\dot{w}}{w+q\xi}. \tag{197}$$

For a differentiable machine or interior mixed allocation under the same fixed resources,

$$\frac{\ddot{A}}{\dot{A}} = (\epsilon_M - \phi)g_A, \qquad \frac{\dot{g}_A}{g_A} = (\epsilon_M - \phi - 1)g_A. \tag{198}$$

In the human expression, evolving expertise and professional demand enter through the wage, while the mixed and machine expressions use the common machine packet price.

When $B_R(t), q(t)$ are positive and differentiable, the corresponding exact identities become

$$\begin{aligned} \frac{\dot{g}_A}{g_A} &= \frac{\dot{B}_R}{B_R} + (\epsilon_H - \phi - 1)g_A - \frac{\dot{w} + \xi\dot{q}}{w + q\xi} && \text{in the human region,} \\ \frac{\dot{g}_A}{g_A} &= \frac{\dot{B}_R}{B_R} - \frac{\dot{q}}{q} + (\epsilon_M - \phi - 1)g_A && \text{in the machine or mixed region.} \end{aligned} \tag{199}$$

Each follows by differentiating $\log \dot{A} = \log B_R - \phi \log A - \log k_*$ within its active regime and subtracting g_A.

J.7 Absorption

To distinguish gradual uptake from a fixed minimum duration, we first allow effective research capability r to approach the published frontier continuously. At a fixed research resource scale,

$$\dot{A} = kr^\eta A^{-\phi}, \qquad \dot{r} = \frac{A - r}{\ell}, \qquad A(0) = r(0) = A_0 > 0, \tag{200}$$

where $k, \ell > 0$ and $\eta, \phi \geq 0$.

Proposition J.10 (Positive absorption time)**.** *System* (200) *has finite-time divergence exactly when $\eta > \phi + 1$. For $\eta < \phi + 1$, it is global and satisfies*

$$\frac{r(t)}{A(t)} \longrightarrow 1, \qquad A(t)^{1+\phi-\eta} \sim (1 + \phi - \eta)kt. \tag{201}$$

For $\eta = \phi + 1$, proportional growth converges to the unique positive solution g_* of

$$g_*(1 + \ell g_*)^\eta = k, \tag{202}$$

and $A(t) \sim Ce^{g_* t}$ for some $C > 0$.

Proof. Local existence and uniqueness hold in the positive quadrant because its vector field is locally Lipschitz. Positivity and the lag formula

$$r(t) = e^{-t/\ell}A_0 + \frac{1}{\ell}\int_0^t e^{-(t-s)/\ell}A(s)\,\mathrm{d}s$$

give $0 < r(t) \leq A(t)$, $\dot{r} \geq 0$, and $\dot{A} > 0$. Hence $\dot{A} \leq kA^{\eta-\phi}$, giving global finite values for $\eta \leq \phi + 1$. On any hypothetical global solution,

$$A(t)^{\phi+1} \geq A_0^{\phi+1} + (\phi + 1)kA_0^\eta t$$

implies $A(t) \to \infty$, and its lag formula then gives $r(t) \to \infty$.

Below the threshold, $0 < g_A \leq kA^{\eta-\phi-1} \to 0$. Writing $u = r/A$ gives

$$\dot{u} = \frac{1 - u}{\ell} - ug_A, \qquad 0 < u \leq 1.$$

For any $\varepsilon > 0$, sufficiently late dates have $g_A \leq \varepsilon$. Comparison with

$$\dot{v} = 1/\ell - (1/\ell + \varepsilon)v$$

gives $\liminf u \geq 1/(1+\ell\varepsilon)$. Together with $u \leq 1$, this proves $u \longrightarrow 1$. With $\gamma = 1+\phi-\eta > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}A^{\gamma} = \gamma k u^{\eta} \longrightarrow \gamma k,$$

and integration proves (201).

At $\eta = \phi + 1$, the ratio obeys

$$\dot{u} = f(u) := \frac{1-u}{\ell} - k u^{\eta+1}.$$

Since $f(0) > 0$, $f(1) < 0$, and $f'(u) = -1/\ell - k(\eta+1)u^{\eta} < 0$, it has one root $u_* \in (0,1)$, and the solution beginning at $u(0) = 1$ decreases to it. By the mean-value theorem,

$$0 \leq u(t) - u_* \leq (1-u_*)e^{-t/\ell}.$$

Thus $g_A = k u^{\eta} \longrightarrow g_* = k u_*^{\eta}$, while the root identity gives $u_* = 1/(1+\ell g_*)$ and (202). Its left side strictly increases from zero to infinity, proving uniqueness. Exponential convergence of u on $[u_*, 1]$ gives integrability of $g_A - g_*$, so $\log A(t) - g_* t$ has a finite limit and $A(t) \sim C e^{g_* t}$.

Above the threshold, a hypothetical global solution has both coordinates tending to infinity as already proved. We define

$$p = \frac{\eta+1}{\eta-\phi-1} > 1, \qquad C = D[1+\ell(p-1)],$$

and choose $D > 0$ sufficiently large that $pC^{1+\phi} \leq kD^{\eta}$, which is possible since $\eta > 1+\phi$. For $0 \leq s < 1$, the functions

$$\underline{A}(s) = C(1-s)^{-p}, \qquad \underline{r}(s) = D(1-s)^{-(p-1)}$$

obey

$$\dot{\underline{r}} = (p-1)D(1-s)^{-p} \leq \frac{\underline{A} - \underline{r}}{\ell},$$

because $C - D = \ell(p-1)D$. Using the exponent identity

$$\eta(p-1) - \phi p = p + 1$$

with the coefficient choice gives

$$\dot{\underline{A}} \leq k \underline{r}^{\eta} \underline{A}^{-\phi}.$$

Both cross derivatives of the original system are nonnegative, so comparison preserves the coordinatewise order between a solution and this subsolution. At contact in either coordinate, its derivative increases with the other coordinate, which proves this comparison by the first-contact argument. Under global existence, the solution eventually exceeds (C, D) and then dominates the divergent subsolution over the following unit interval. Such domination contradicts finite values on that compact interval, proving a finite maximal endpoint. Bounded A there would bound $r \leq A$ and permit continuation by local Lipschitzness, so A diverges. Bounded r would instead bound $\dot{A} \leq k r^{\eta} A_0^{-\phi}$, so r diverges as well. □

J.8 Research duration

With a minimum research duration, every advance waits before contributing to another completed research cycle. For $k, \tau > 0$, $\eta, \phi \geq 0$, and continuous positive history on $[-\tau, 0]$, the frontier follows

$$\dot{A}(t) = k A(t-\tau)^{\eta} A(t)^{-\phi}. \tag{203}$$

Our next result gives its growth order as well as its finite-date existence. Transforming to a delayed power equation connects this law to the functional differential equations studied by Appleby et al. (2013), while the proof below derives the three orders directly for our research specification.

Proposition J.11 (A positive minimum research duration)**.** *Equation* (203) *has a unique positive solution that remains finite at every finite date. Below the threshold $\eta < 1 + \phi$, it satisfies*

$$A(t)^{1+\phi-\eta} \sim (1 + \phi - \eta)kt. \tag{204}$$

At $\eta = 1 + \phi$, positive constants C_-, C_+ satisfy

$$C_- e^{gt} \le A(t) \le C_+ e^{gt}, \qquad g e^{(1+\phi)g\tau} = k, \tag{205}$$

where $g > 0$ is the unique solution, and $\log A(t)/t \to g$. Above the threshold $\eta > 1 + \phi$,

$$\lim_{t\to\infty} \frac{\log \log A(t)}{t} = \frac{1}{\tau} \log \frac{\eta}{1+\phi}. \tag{206}$$

Proof. On $[0, \tau]$, delayed capability is a known continuous bounded positive function. Direct integration gives

$$A(t)^{1+\phi} = A(0)^{1+\phi} + (1 + \phi)k \int_0^t A(s - \tau)^\eta \, \mathrm{d}s,$$

whose positive right side determines a unique finite solution. On each successive interval $[n\tau, (n + 1)\tau]$, the previous interval supplies known bounded delayed inputs and gives

$$A(t)^{1+\phi} = A(n\tau)^{1+\phi} + (1 + \phi)k \int_{n\tau}^t A(s - \tau)^\eta \, \mathrm{d}s.$$

Induction gives a unique positive solution at every finite date.

For the asymptotic orders, set $X = A^{1+\phi}$, $a = (1 + \phi)k$, and $r = \eta/(1 + \phi)$, giving

$$X'(t) = aX(t - \tau)^r. \tag{207}$$

It is increasing for $t \ge 0$, and the positive minimum of its initial history bounds its derivative below by a positive constant. Hence $X(t) \to \infty$.

For $r < 1$ and $t \ge \tau$,

$$0 < \frac{X'(t)}{X(t)} \le aX(t)^{r-1} \longrightarrow 0.$$

Integration over an interval of length τ yields $X(t - \tau)/X(t) \to 1$. Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t} X^{1-r} = (1 - r)a \left[\frac{X(t - \tau)}{X(t)}\right]^r \longrightarrow (1 - r)a,$$

which proves (204).

For $r = 1$, the positive root of $\lambda = ae^{-\lambda\tau}$ exists and is unique because $\lambda e^{\lambda\tau}$ strictly increases from zero to infinity. Every function $Ce^{\lambda t}$, $C > 0$, solves (207). Taking the positive minimum and finite maximum of $X(s)e^{-\lambda s}$ over its initial history provides lower and upper such solutions. Comparison on successive delay intervals preserves their order at all later dates. Taking $(1 + \phi)$-th roots gives (205) with $g = \lambda/(1 + \phi)$, and taking logarithms gives the limiting average growth rate.

For $r > 1$, $Y = \log X$ is positive and unbounded after a finite date and satisfies

$$Y'(t) = a \exp\{rY(t - \tau) - Y(t)\}.$$

We choose positive $\lambda_- < \log r/\tau < \lambda_+$. For $Z(t) = Ce^{\lambda t}$, the right side evaluated on Z becomes

$$a \exp\{(re^{-\lambda\tau} - 1)Z(t)\}.$$

At $\lambda = \lambda_-$, its exponent coefficient is positive, so a sufficiently large Z_0 satisfies

$$\lambda_- z \le a \exp\{(re^{-\lambda_- \tau} - 1)z\} \qquad (z \ge Z_0).$$

We choose t_0 sufficiently late that $Y(s) \ge Z_0$ throughout $[t_0 - \tau, t_0]$. Then $Z_-(t) = Z_0 e^{\lambda_-(t-t_0)}$ lies below this history segment and is a subsolution from t_0 onward.

At $\lambda = \lambda_+$, the exponent coefficient is negative, so the evaluated right side is at most a. We choose $C \ge a/\lambda_+$ sufficiently large that $Z_+(t) = Ce^{\lambda_+(t-t_0)}$ lies above the same history segment. Its derivative is at least a from t_0 onward, giving a supersolution. At a first contact, the current values coincide and ordered delayed values give ordered derivatives, because the delayed argument enters monotonically. Hence comparison gives $Z_-(t) \le Y(t) \le Z_+(t)$. Taking logarithms and dividing by t bounds the limit inferior and limit superior of $\log Y(t)/t$ by λ_- and λ_+. Bringing both constants to $\log r/\tau$ proves that limit. Finally,

$$\log \log A(t) = \log Y(t) - \log(1 + \phi),$$

which proves (206). □

Above the threshold, the limit describes a double-exponential logarithmic order, with the speed determined by the productivity ratio $\eta/(1+\phi)$ and the minimum cycle duration τ. Gradual absorption continuously assigns positive weight to recent improvements, whereas a fixed minimum duration passes each advance into a later cycle. These distinct timing requirements preserve different amounts of the research feedback.

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